MAPÚA MALAYAN COLLEGES MINDANAO

Geospatial Causal Inference for Barangay-Level Maternal Care Assessment: A Causal Forest Framework on Antenatal Intervention Outcomes in Davao City

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A Thesis Report Submitted to the College of Computer and Information Science

in Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Computer Science

Mapúa Malayan Colleges Mindanao

Davao City

May 2025

APPROVAL SHEET

The thesis, entitled **"Geospatial Causal Inference for Barangay-Level Maternal Care Assessment: A Causal Forest Framework on Antenatal Intervention Outcomes in Davao City"** prepared and submitted by Group 25-CS-006 consisted of **Raphael Mico C. Barrios, JC-Rey V. Cata-al** & **Human Walter R. Olo** in partial fulfillment of the requirements for the degree of **Bachelor of Science in Computer Science** is hereby accepted.

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DEAN

ACKNOWLEDGEMENT

First, we would like to express our gratitude and dedicate this work to our parents, families, and friends for their moral support, love, and patience throughout our academic journey. Their encouragement has been the greatest motivation to preserve and complete this endeavour.

We would also like to express our sincerest gratitude to Dr. Patrick D. Cerna, DIT, our thesis adviser, for his valuable guidance, patience, and expertise throughout the processes of this study. His insightful and valuable advice and encouragement have been vital in the successful completion of our research.

We extend our gratitude and heartfelt appreciation to our defense panel members, Sir Martzel Baste, Dr. Genevieve Pilongo, and Sir Clyde Chester Balaman, for their constructive feedback, and valuable recommendations given when approached, which have strengthened the quality and clarity of this thesis.

Our sincerest thanks and appreciation go to the Davao City Health Office (CHO) for granting us access to the Routine Health Information System (RHIS) data and for their kind cooperation. Their assistance was essential to ensure that this research is feasible and contribute meaningfully to public health decision-making.

Finally, we also thank the College of Computer and Information Science (CCIS) of Mapua Malayan Colleges Mindanao for providing the facilities and academic support that made this study possible.

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LIST OF ABBREVIATIONS

AI Artificial Intelligence

ANC Antenatal Care

ATE Average Treatment Effect

BLP, Best Linear Projection

CATE, Conditional Average Treatment Effect

CBC, Complete Blood Count

CHO, City Health Office

CML, Causal Machine Learning

CVD, Cardiovascular Disease

DML, Double Machine Learning

DOH, Department of Health

DSA, Data Sharing Agreement

GDM, Gestational Diabetes Mellitus

GIS, Geographic Information System

HTE, Heterogeneous Treatment Effect

MCH, Maternal and Child Health

ML, Machine Learning

MMR, Maternal Mortality Ratio

PNC, Postnatal Care

PSA, Philippine Statistics Authority

QGIS, Quantum Geographic Information System

RHIS, Routine Health Information System

UHC, Universal Health Coverage

WHO, World Health Organization

Article 1

Geospatial Causal Inference for Barangay-Level Maternal Care Assessment: A Causal Forest Framework on Antenatal Intervention Outcomes in Davao City

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1. Introduction

Maternal health refers to the health of women during pregnancy, childbirth and the postnatal period (WHO, 2019). All three stages of pregnancy are critical so ensuring positive experiences at every stage is important for both maternal and newborn survival and well-being. Maternal health care and outcomes have improved globally over the recent decades, however maternal mortality remains alarmingly high, particularly in low- and middle-income countries. Maternal morbidity refers to those occurring during pregnancy, childbirth, or within the postpartum period. According to the World Health Organization, in 2020, there was a record of an estimated of 287,000 women died from pregnancy-related causes, many of which, could have been prevent with timely and appropriate healthcare. The leading causes of these maternal mortalities include direct causes such as severe bleeding, infections, high blood pressure during pregnancy, unsafe abortion, and obstructed labor (WHO, 2025).

In the Philippines, maternal health remains a major public health concern. Despite national programs like the Maternal, Neonatal, and Child Health and Nutrition (MNCHN) Strategy, maternal mortality rates continue to challenge the health system of the country. The Philippine Statistics Authority (PSA) reports that the maternal mortality ratio (MMR) in the Philippines remains above the SDG 3 target of 70 per 100,000 live births, with significant disparities between urban and rural areas. According to Inquirer (2023), the PSA actually reported that there were 2,478 women who died from maternal causes in 2021, the MMR for that is 189.21 per 100,000 live births which is the third deadliest number of maternal mortality next to 1951 and 1952.

In Davao City, maternal health is still a multifaceted issue, particularly in inequalities in the quality and accessibility of maternal care across barangays. In 2021, the City Health Office (CHO) recorded 22 maternal mortalities, which were primarily caused by hemorrhage

and eclampsia, both are preventable with timely and quality prenatal care (City Government of Davao, 2022).

A qualitative study conducted by Cagayan et al. (2022), in rural health units in Luzon revealed the systemic challenges faced by mothers in accessing maternal health services. The study reported the challenges that included long travel distances, lack of transportation, poor facility readiness, and intermittent availability of skilled health professionals. Many women also reported feeling disrespected or neglected during health visits, which discouraged regular check-ups and facility-based deliveries. These insights highlight the persistent socio-structural and accessibility barriers that limit the effectiveness of existing maternal health programs. The study also shows the importance of local health units providing maternal care, especially the need for capacity building, resource allocation, and localized decisions-making tools to better help rural communities.

In response to these recorded maternal mortalities, the CHO intensified their efforts through the Safe Motherhood Program, having multiple initiatives aimed at improving maternal health services. This program includes services such as 18 batches of consultive meetings for district health staff, District Facebook Pages for pregnant women to receive necessary prenatal updates and advocacy materials, a planned Safe Motherhood Hotline and, a mobile clinic services that aims to bring prenatal care, nutrition counseling, laboratory tests, and micronutrient supplementation directly to communities.

The barangay and district health centers currently provide prenatal services that include nutritional assessments via BMI, Tetanus-Diphtheria immunization, micronutrient supplementation specifically iron and folic acid, deworming, STI and laboratory testing, and gestational diabetes screening. Despite these efforts, a study by Matutino et al. (2024) revealed inconsistent perceptions of care quality among mothers in selected barangays, citing resource limitations, follow-up gaps, and variable patient experiences as ongoing challenges.

While the Davao City Health Office (CHO) collects extensive data on maternal health and outcomes, there is still a critical analytical gap that persists, and it is the inability to move from identifying correlational patterns and disparities to estimating the causal effectiveness of specific interventions across the different barangays. Prior studies in the Philippines have mapped out the maternal health challenges, such as identifying systemic barriers making it hard for mothers to access help (Cagayan et al., 2022) and inconsistent perceptions of car quality in Davao City barangays (Matutino et al., 2024). Similarly, predictive machine learning models, like the one Macrohon et al. (2022) has developed for risk classification, just focuses on forecasting outcomes but do not estimate the causal impact of maternal health interventions. This leaves a fundamental question unanswered: "Which specific maternal health interventions have the strongest causal impact on live birth rates in different barangays of Davao City?". This study seeks to uncover the causal mechanism that link antenatal interventions to live birth rates. We specifically investigate how barangay-level factors like healthcare worker density, facility access, and population demographics, can moderate the effect of an intervention. As an example, "Does increasing the rate of facility-based deliveries causally increase live birth rates more significantly in remote barangays with low midwife-to-population ratios than in urban centers?". By formally estimating these heterogenous causal effects, this research aims to provide an evidence-based foundation for targeted resource allocation, moving beyond the one-size-fits-all policy recommendations.

These disparities despite the current efforts emphasize the need for a data-driven approach to support evidence-based decision-making in maternal care. While these services are in place, there remains a gap in the ability to assess the effectiveness of maternal care interventions and identify areas that most need support. The absence of these tools makes it difficult to track the success of interventions across the Davao City barangays over time.

To address this, this study will begin with the collation of selected barangay-level data from Davao City's Routine Health Information System (RHIS). Although RHIS is an administrative dataset, in this study it is considered the primary dataset because it was obtained directly from the City Health Office (CHO) before any cleaning or processing. This step is data gathering of the relevant indicators of this study such as, antenatal care services, maternal outcomes, and interventions activities across the different health centers in Davao City. Consolidating these varied datasets into a structured and analyzable format, this study will be able to have a more granular analysis and make sure that both successful and underserved areas are properly captured.

Causal forest is a machine learning method that focuses more on the identification of heterogeneous causal effects of treatment and is unlike the conventional machine learning methods (Kakimoto et al., 2022). In recent years, causal forests have been applied in various studies such as economics, transportation, and agriculture to make site-specific or context-specific decisions. A study by Kakimoto et al. (2022) utilizes the causal forest approach in the agricultural context, specifically to generate economically optimal site-specific input management recommendations based on the gathered on-farm experimentation data. Similarly, in the medical field, a study by Raghavan et al. (2022) utilizes causal forest to explore whether tighter sugar control helped some diabetes patients more than others. They found differences between patient groups but also saw that those patterns did not always repeat in other similar studies. These studies mentioned show the strength of causal forest in giving insights on how treatment effects vary across different population subgroups. This capability is relevant to this study's focus on barangay-level variation in maternal health outcomes.

Two methods were taken into consideration to tackle this study, and while Causal Forest was selected by the researchers due to it having a distinct ability to estimate Conditional Average Treatment Effects (CATEs) at a granular level, another method was also considered

which is the Double Machine Learning (DML). DML is a statistically strong and effective way for estimating average treatment effects, but is it not designed to capture the heterogeneity of effects across diverse settings. In contrast, Causal Forest can reveal how maternal health interventions perform differently depending on barangay specific factors like access to facilities, resource allocation, and personnel distribution. A study by Zheng and Yin (2023) explained how Causal Forest is particularly well-suited for observational studies, because it enables nuanced, interpretable causal analysis. Furthermore, another study by Credit and Lehnert (2023) highlighted that Causal Forest is especially effective in spatially distributed datasets, where treatment impacts are uneven across the geography, this is an ideal choice for this study's target of health disparities within Davao City's 182 barangays. These strengths from this selected model align closely with the goal of the study which is to generate localized, actionable insights for maternal healthcare planning and policy.

To ensure the reliability of the hybrid model's output, the study also includes an evaluation phase using both cross-validation and falsification tests. In geospatial datasets such as maternal health indicators across barangays, using standard random cross-validation can lead to an overestimated model performance due to spatial autocorrelation, this is the tendency for neighboring areas to exhibit similar characteristics. To address this, this study applies spatial cross-validation, which partitions the data based on geographic blocks, making sure that the training and test sets are spatially independent. A study by Sun et al. (2023) has emphasized the important role of Spatial Cross-Validation in GeoAI applications, also showing that the traditional Cross-Validation techniques can produce overly optimistic results in spatial domains. Along with the Cross-Validation strategy, falsification tests will also be employed to evaluate the causal validity of the model. These tests detect potential confounding or misspecification by applying the model to outcomes that should not be influenced by maternal interventions, such as health indicators not related to maternal health. A study from Liu et al.

(2023) highlighted that assumption-lean falsification tests are vital when deploying double-robust machine learning estimators in any observational settings, making sure of credible causal inferences by identifying the false positives that may arise from unobserved biases.

Together with the hybrid model, this study also integrates a web-based platform as a visualization layer rather than a core contribution. Built with React with Vite for the frontend and Flask for the backend, the system allows users to interact with heatmaps representing each barangay in Davao City. Color gradients to indicate the severity or effectiveness of the maternal health interventions will be shown on these heatmaps. This is done to enhance the accessibility and the interpretability of the study's results.

This study aims to address the gaps of maternal health care in the Davao City barangays with the help of causal forest and geographic heterogeneity. This study proposes a data-driven system that integrates routinely collected and selected health data and applies causal inference and geospatial analysis. With this combination of statistical modeling and spatial mapping the system will help the workers and policymakers of the health sector to see and understand which interventions work effectively, and in which barangays. These insights generated by the hybrid model will serve as a valuable tool for evidence-based planning for resource planning and targeted response in district to barangay-level maternal healthcare efforts.

Specifically, this study aims to:

1. To construct a collated dataset by consolidating data on antenatal care services, maternal health outcomes, and interventions from the records of the Davao City Routine Health Information System (RHIS)
2. To train a hybrid model that applies Causal Forest algorithm to estimate how the effects of maternal health interventions vary across barangay-level subgroups and incorporate geospatial heterogeneity to visualize these heterogeneous treatment effects.

3. To evaluate the maternal care interventions hybrid model using Spatial Cross-Validation and Falsification Tests.

2. Related Works

This section reviews other studies that provide an overview of the study which focuses on the essence of maternal care, machine learning applications that forecast diseases based on its performance metrics, and the integration of data-driven models to support early intervention, risk assessment, and clinical decision-making in maternal healthcare. With these, the researchers aim to identify the gaps in other studies, extract relevant parameters, and adopt techniques in gathering and developing data for this thesis.

2.1 Maternal Care

According to the World Health Organization (WHO), maternal health refers to the wellbeing of women during pregnancy (antenatal), childbirth (intrapartum), and the postnatal period, where each stage should be considered as a positive experience, ensuring that women and their babies reach their full potential for health and well-being.

Antenatal care (ANC) refers to healthcare provided from the start of pregnancy until birth. It also includes preparing for physical and mental well-being by receiving advice from an obstetrician regarding practices to ensure a safe pregnancy. The components of ANC include risk identification, prevention and management of pregnancy-related or concurrent diseases, and health education and promotion (WHO, 2016).

Intrapartum is defined as the time spanning childbirth, from the onset of labor through delivery of the placenta. It can also be referred to both the woman and the fetus (Clinicalinfo, 2021). During childbirth, it is divided into 3 stages. The first stage begins with the initiation of labor and continues until the cervix is fully dilated to 10cm. The second stage spans from full cervical dilation to the delivery of the baby. The third stage occurs after the baby is delivered and concludes when the placenta is on full expulsion (Hutchison *et al*., 2025).

Postnatal care (PNC) refers to the care given to the mother and newborn immediately after childbirth. This period usually lasts 6 to 8 weeks and ends when the mother's body has nearly returned to its pre-regnant state (Lopez-Gonzalez & Kopparapu, 2022). This stage includes monitoring the mother for postpartum bleeding, infection, and mental health issues, while also providing newborn assessments, breastfeeding support, immunizations, and family planning services (WHO, 2022; UNICEF, 2021). It is essential to ensure recovery and long-term health for both the mother and the baby. Lopez-Gonzalez and Kopparapu (2022) states that there are components in PNC, namely vaginal pain, vaginal bleeding/discharge, breastfeeding, nutrition and exercise, breast engorgement, bladder and bowel function, sexual relations, contraception, education, and miscarriage, stillbirth, or neonatal death.

2.2 Maternal Care Diseases

Maternal care diseases pose significant risks to women's health during pregnancy, childbirth, and even the postnatal period. According to the World Health Organization (WHO), maternal mortality is predominantly preventable, it is often caused by severe bleeding (postpartum hemorrhage), infections found usually after childbirth, pre-eclampsia and eclampsia during pregnancy, complications from delivery and unsafe abortion practices. Approximately, 287,000 women die annually due to these complications. There are also indirect causes such as anemia, HIV/AIDS, malaria, and cardiovascular disease. These diseases contribute to poor maternal outcomes.

In addition, intrauterine fetal demise (IUFD) is also a significant concern in maternal healthcare. A cross-sectional study done by Tantengco et al. (2024) and conducted at the Philippine General Hospital from 2012 to 2021 analyzed that second-trimester and third-trimester IUFD cases to understand their epidemiology and placental pathology. Hypertensive disorders of pregnancy were among the leading maternal conditions associated with IUFD as

the study has found. Placental pathologies, such as infarctions and abruptio placentae, were also commonly observed in these cases. The findings from this study highlight the need for and importance of early detection and management of hypertensive disorders and placental abnormalities to prevent IUFD and improve the maternal outcomes.

In pregnancy, there is a hypertensive disorder called preeclampsia that is characterized by high blood pressure and signs of damage to the liver and kidneys, if left untreated, can lead to life-threatening complications for both the mother and the fetus. A study done in the Philippine population emphasized the clinical significance of preeclampsia by examining its underlying biological factors. Amosco et al. (2019) highlighted the preeclampsia in Filipino women may be influenced by molecular-level disruptions, such as involving the ACVR2A gene, which regulates placental and vascular function. This study points to the disease's complexity and the need for comprehensive prenatal screening to help prevent and reduce associated maternal and neonatal complications.

2.3 Maternal Care Programs

Strategies and actions are implemented to improve the well-being of pregnant women. The strategies that will be later stated in this section will serve as potential indicators to the thesis project. However, it is important to note that these health indicators differ across countries, regions, and socioeconomic levels (Haakenstad *et al.*, 2022).

There are various governments that have developed frameworks that promote the welfare of maternal care. For instance, the Partnership for Maternal, Newborn & Child Health (PMNCH), a subsidiary of WHO, created a strategy that seeks to reduce maternal, newborn, and child mortality and morbidity, including stillbirths. The strategy requires measurable policy, financing, and service delivery improvements by countries, including improved national policies, improved country health expenditure per capita, and improved coverage of

essential services (PMNCH, n.d). The PMNCH has a list of indicators that showcase its commitment to their strategy, but only certain subdomains are considered and will be used in the context of this study, as shown below.

Table 1: *Maternal, Newborn and Child Health Commitments*

Sub-domains	Indicators
Universal access to health services for Women, Children, and Adolescents' Health (WCAH), including stillbirths	Antenatal, childbirth, and postnatal care, including emergency obstetric and newborn care
	Births attended by skilled health personnel
	Free access to health services for pregnant women and newborns
	Universal access to health services for children
	Essential MNCH services included in UHC packages
	Children fully immunized
	Proportion of target populations covered by all vaccines included in national programme
	Stillbirth prevention (Global Strategy stillbirth rate)
Health system strengthening to promote MNCH including for stillbirths	National policies/laws to review maternal deaths, stillbirths and neonatal deaths
	Health worker density and distribution including midwifery and nursing
	Proportion of health facilities that have a core set of relevant essential medicines available and affordable on a sustainable basis

2.4 Data Acquisition and Preprocessing

The foundational components of developing an accurate and timely epidemiological model are effective data acquisition and preprocessing.

One notable approach made by Lijović et al. (2024) involves the automated extraction of electronic health record (EHR) data from intensive care units (ICUs). This study assessed the feasibility of such automated extraction of data during the COVID-19 pandemic and has demonstrated that the automated EHR data extraction could accelerate clinical insights by providing timely and structured data for analysis.

In addition, the integration of exogenous variables into epidemiological models has been explored to improve forecasting accuracy. A hybrid model combining data-driven neural ordinary differential equations with traditional SEIR compartment models has been proposed. According to Ziarelli et al. (2024), the framework of this model allows the incorporation of external factors, such as meteorological data, into the modeling process and enables a more nuanced understanding of transmission dynamics.

Water-based surveillance is also another innovative data source, utilizing wastewater data in epidemiological models could forecast COVID-19 outbreaks in rural communities with a lead time of 9 to 15 days (Meadows et al., 2025). This approach offers a non-invasive and cost-effective means of monitoring community-level infection trends.

All these studies emphasize the critical role of having effective data acquisition and preprocessing methods in enhancing the responsiveness and accuracy of epidemiological models.

2.5 Related Geospatial Analysis Studies

Geospatial analysis utilizes geographic or spatial information to understand and analyze patterns, relationships, and trends in data (Charles *et al.* 2024). Several studies have utilized this method to examine health risks and outcomes in maternal care and other health-related contexts.

Rubaiya *et al.*'s study revolves around using spatial analysis to identify low birth weights. They further used spatial dependence modeling, namely fixed-effects, mixed-effects, and spatial-lag to account for geographic autocorrelation among districts in Bangladesh. Using this method, hotspots are displayed on choropleth maps showing predicted low birth weight cases on the district scale.

Yaya *et al.*'s study uses spatial and Bayesian networks to determine Maternal Mortality Ratio (MMR) found in 54 African countries. This study showed clusters from the spatial analyses where seven countries within the Middle and West Africa are considered as hotspots, while countries like South Africa and Namibia, as well as eight other countries in Africa, formed low-high clusters which indicate that they have significantly low MMR but within those neighborhoods of countries with significantly high MMR.

Liu *et al.*'s (2024) study uses sequence mining as their algorithm to discover and visualize common patterns from their extracted dataset, focusing primarily on prenatal care concerns in relation to urban built environment factors where the main outcome is postpartum depression. Their results discover that three common types where women experience with the highest rates of PPD reside in neighboring areas in terms of walkability and street connectivity, and air quality.

Rajuroy and Emmanuel's (2025) study focuses more on exploring the application of machine learning techniques to develop predictive models for mapping maternal and child health disparities alongside using geospatial data. The study shows that machine learning algorithms can be applied within geospatial information systems to perform advanced spatial analyses and make predictions about MCH disparities, in which it can predict health outcomes, identify vulnerable areas, and improve decision-making.

Furthermore, the following section delves deeper into other related studies that utilize machine learning algorithms applied in the context of maternal health research.

2.6. Applicable Machine Learning Models

There are numerous studies that have applied various machine learning models to predict health risks and outcomes in maternal care. Most of the related studies combined a number of algorithms to create a ensemble ML model to gather accurate predictors.

For instance, Sani & Ahmed's (2023) study used Random Forest and XGBoost to effectively highlight significant demographic and healthcare access-related predictors. Khadidos *et al.* and Togunwa *et al.* utilized ensemble and deep hybrid models, creating their own models to achieve accurate results.

Khadidos *et al.* study used a mixture of Decision Trees, Gradiant Boosted Tree, Random Forest, and K-Nearest Neighbors (KNN) to create a Large Language Model named the Quad-Ensemble Machine Learning Framework, while Togunwa *et al.* used Artificial Neural Networks (ANN) and Random Forest.

Kumar *et al.* used Logistic Regression (generalized linear model), Support Vector Machine (linear support vector classification), CatBoost gradient boosting (tree-based), and Artificial Neural Network (multilayer perception).

Macrohon *et al.* used a combination of Decision Trees, Random Forest, Support Vector Machine, K-Nearest Neighbors, Naive Bayes, and Multilayer Perception through hyperparameter tuning.

2.7 Model Evaluation and Performance Metrics

Developing Machine Learning models require evaluation based on their performance to ensure accuracy, efficiency, and reliability in real-world applications, providing insights to make accurate predictions and balance errors. Performance metrics such as accuracy, precision, and recall are used in maternal health risk prediction models such as the studies of Bertini *et al.* (2022) and Sani, J., & Ahmed, M. (2023), in which they both use these to ensure effective

classification, minimize errors, and balance metrics to avoid unnecessary interventions. Other metrics, from the studies of Khadidos *et al.* (2023) and Togunwa *et al.* (2022) use F1 score which combines the output of recall and precision values, and Receiver Operating Characteristic Area Under the Curve (ROC-AUC) to evaluate the performance of binary classification models using the True Positive Rate (TPR) against the False Positive Rate (FPR).

Having identified the various Machine Learning models applied in the field of maternal care, it is essential how these models performed based on the evaluation metrics. Sani & Ahmed's study concludes that Random Forest achieved the highest performance, with an accuracy of 70%, precision of 0.69, recall of 0.71, and AUROC of 0.74, followed by XGBoost with an accuracy of 69% and AUROC of 0.72. Khadidos *et al.'s* study concluded that Gradient Boosted Trees (GBT) with ensemble stacking had the best overall performance, achieving 90% accuracy for the "High Risk" class and 0.86 F1-score across all classes (High, Medium, Low). Togunwa *et al.*'s study stated that their MaternalNET-RF model, consisting of both ANN and RF, performs best when compared to other models, with an accuracy score of 0.9488, and its precision, recall, and F1-Score ranged from 0.9300 to 0.9700. Macrohon *et al.* concluded that their modified Decision Tree algorithm performed the best among other algorithms with an accuracy of 93.70%, sensitivity rate of 90.05%, and a specificity rate of 97.37%.

Based on the results given, it shows that hybrid models tend to outperform single-algorithm approaches, like Khadido *et al*'s GBT ensemble and Togunwa *et al*'s MaternalNET-RF model. Sani & Ahmed's Random Forest model reached an AUROC result of 0.74, but it fell behind in precision and overall accuracy compared to Macrohon *et al*'s modified Decision Tree and Togunwa *et al*'s hybrid model.

Despite the promising results, a significant research gap remains: most studies rely on datasets limited to specific populations, making them less applicable to the unique demographic and health behaviors in Davao City. Additionally, their scopes often focus solely on one aspect

of maternal care—either antenatal, intrapartum, or postnatal—rather than the whole range of the process. The closest comparable research is Macrohon et al.'s study, which utilized data from health centers in Daraga, Albay. However, the dataset was restricted to the Pre-Natal Target Client List, lacking variables related to delivery and postnatal outcomes. Furthermore, due to the limited data, the authors employed the Synthetic Data Vault (SDV) to generate additional entries, which may introduce outliers or non-representative data, potentially affecting the accuracy and validity of predictive outcomes.

2.8 Machine Learning Techniques in Maternal Care

Machine Learning models play a critical role in the field of maternal health, ensuring that they are considered transparent, reliable, and ethically sound. Related studies show that there are techniques that potentially improve the performance metrics of data models.

Techniques such as Shapley Additive Explanations (SHAP) and feature importance ranking are increasingly used in maternal health research (Sani & Ahmed, 2023; Al Mashrafi et al., 2022). SHAP is utilized by assigning each feature an importance value for a particular prediction, where the components are the identification of a new class of additive feature importance measures, and theoretical results showing there is a unique solution in this class with a set of desirable properties (Lundberg, S. & Lee, S., 2017). Macrohon *et al.* used the Synthetic Minority Oversampling Technique (SMOTE) to balance data. SMOTE samples data from the minority class by simply generating data points on the line segment connecting a randomly selected data point and one of its K-nearest neighbors (Elreedy, D., & Atiya, A., 2019). Khalidos *et al.* used ensemble methods like Bagging, Adaptive Boosting (AdaBoost), Stacking, and Voting.

Bagging is a method for generating multiple versions of a predictor and using these to get an aggregated predictor, which averages over the versions when predicting a numerical

outcome and does a plurality vote when predicting a class (Breiman, 1996). AdaBoost combines multiple week classifiers to create a strong classifier by sequentially adding classifiers to correct the errors made by previous models giving more weight to the misclassified data points. Stacking explores a space of different models for the same problems, where the idea is to solve a learning problem with different types of models that are capable to learn some part of the problem, but not the whole scope of the problem. Voting considers each base learner's prediction for a given data instance and outputs a final prediction determined by whatever the majority of learners predict (IBM, 2024). Al Mashrafi, S., Takafori, L., & Abdollahian, M.'s study utilized Principal Component Analysis (PCA) which reduces the number of dimensions in large datasets to principal components that retain most of the original information by transforming potentially correlated variables into a smaller set of variables called principal components (IBM, 2023).

These techniques not only enhance performance but also ensure transparency and trust where interpretability and fairness are essential.

2.9 Related Causal Forest Studies

Unlike most machine learning models, Causal Machine Learning (CML) focuses on identifying causal impacts of an event (Kakimoto *et al.,* 2022). Causal forests, an example of CML, is an ensemble tree-based learning method which discovers and estimates heterogeneous treatment effects using a data-driven approach (Jawadekar *et al.*, 2023). Despite its recent creation, numerous studies have proven that causal forests are effective in predicting certain indicators from economics and transportation to agriculture and medicine uses.

Liu, T., Liu, J., and Cheng, T. (2023) evaluated the heterogeneity of green credit policy on labor investment efficiency by analyzing Chinese listed firms from 2005 to 2020 using the difference-in-differences method and causal forest approach. Their results show that the green

credit policy improved labor investment efficiency in high-polluting firms by reducing over-investment. Furthermore, the usage of causal forests revealed nonlinear and heterogeneous treatment effects, where their indicators (lower market capitalization, lower liquidity, and higher leverage experience) show the declination of firms due to financial constraints.

Guo *et al.* used Causal Forest to predict the duration of traffic incidents, conducting a comprehensive causal analysis of traffic incident duration using historical data and including different types of roads across Tianjin, China. The framework they have implemented enables the assessment of Average Treatment Effects (ATEs) of various factors on incident duration. The indicators that were analyzed were policies related to lane configurations are explored, which further concludes the necessity of considering causal effects in traffic management decisions.

Wu, Z., Zhang, Y., and Zhang, M.'s (2023) study focuses on China's Minimum Living Standard Guarantee subsidy on household transportation expenditures, where the program serves as a social safety net, providing individuals to meet their basic needs and escape poverty. Data was collected from the Chinese Social Survey, which is then combined with the Propensity Score Matching (PSM), Causal Forest, and Marginal Treatment Effect processes to estimate the subsidy's effects. The Propensity Score Matching mitigates selection bias by matching the recipients with comparable non-recipients, whereas Causal Forest captures the heterogeneity of treatment effects across various household profiles. The analysis indicates the mediating mechanism that MLSG impacts household transportation expenditures. The results that were provided have important implications for policy formulation, particularly in rural-urban area differences.

Kakimoto *et al.'s* (2021) study uses Causal Forest to estimate site-specific crop yield response to changes to input management (i.e seed, fertilizer) in making economically optimal site-specific input management recommendations. The study uses Monte Carlo simulations to

compare CF against other Machine Learning Methods, in which Causal Forest consistently outperformed the other methods across all modeling scenarios in terms of economically optimal nitrogen rates' prediction and profit generation. Other results, however, show that Causal Forest does not perform better economically than the other traditional methods. The conclusion further states that using yield prediction accuracy should not be used as a model selection criterion if the goal is to create a site-specific contextual map.

Susukida *et al.'s* (2025) study examines treatment effect heterogeneity on Substance Use Disorder treatments, applying Causal Forest to examine in psychology SUD treatments. The data collected came from twelve randomized controlled trials of nine psychosocial treatments from the National Institute on Drug Abuse Clinical Trials Network. With that dataset, their Causal Forest approach estimates the conditional average treatment effect on drug abstinence, followed by an aomnibus test to examine the presence of HTE. Linear project analysis was additionally conducted to examine relationships between baseline covariates and CATE. Despite the lackluster results of the omnibus test, the CATE standard deviation was larger than the magnitude of ATE estimate in three indicators: Motivational Interviewing, Motivational Enhancement Therapy, and Brief Strategic Family Therapy, which suggests that the treatment effect was not uniform in these interventions. The study concludes that in lieu of the model performance, the covariates that were related to treatment effects are noted to be effective. The study further highlights the importance of diverse, larger-scale studies that consider a broader spectrum of baseline covariates.

2.10 Challenges and Limitations in Maternal Health Prediction

Despite the growing applications of machine learning in maternal health, challenges and limitations still exist, which hinders its full potential in clinical and community settings. Data quality and availability are two primary issues. Many studies rely on limited datasets that are

often specific to a particular region or health facility, which not only gives potential bias by using only the local data but also restricts generalizability across different populations like Sani & Ahmed's study that focuses primarily on the 2020 Somalia Demographic and Health Survey (SDHS), and Macrohon *et al.*'s study.

Imbalanced datasets are also a problem where high-risk cases are significantly fewer than low-risk ones, further potentially biasing the model's prediction. Khadidos *et al.*'s study highlights the high-risk patients as the most accurate predictor than medium or low risk patients. Although techniques like SMOTE help address this, they still risk overfitting or creating artificial patterns.

Data handling, model transparency, and system implementation may complicate the use of machine learning models in the context of Davao City. Rural areas limit the ability to gather consistent datasets, such as the study of Macrohon *et al.* that only uses one barangay of a province which limits the integrity of the dataset.

2.10 Ethical, Legal, and Deployment Considerations

When conducting this thesis, it is important to understand the considerations in terms of its ethics, legality, and deployment. In the context of Artificial Intelligence or Machine Learning, according to Wei *et al.*, it is increasingly influencing healthcare, providing transformative opportunities for nursing practice. However, the privacy and security of patient data is a critical ethical concern when used as training for AI models, which will lead to potential biases like data bias, development bias, and interaction bias (Hanna *et al.,* 2025). To improve this problem, it is important to understand fairness, accountability, transparency, and ethics in the field of not just maternal care but healthcare in general. He *et al.* keyed out these four aspects as the primary focuses is on patients' well-being and safety. Fairness holds paramount significance in guaranteeing that AI does not worsen societal disparities; Accountability ensures that

individuals responsible for the implementation of AI can be held answerable for their decisions; Transparency assumes that AI remains open to inspection and compliant to audits for possible biases or inaccuracies; Ethics guarantees that AI is constructed and utilized in manners that align with prevailing social values and norms (He *et al.*, 2025).

In the context of this thesis, this program will be deployed to the City Health Office of Davao City, and it is important to note the usage of datasets from the said department must be protected, secured, and privatized under the Republic Act 10173 or the Data Privacy Act of 2012 (OfficialGazette.gov.ph, 2012). Upon app deployment, the users must evaluate first the system in a real-world environment, to confirm whether or not it meets the intended requirements and effective functions of the program. As an example, one of the methods to use is black box testing, where it involves testing a system with no prior knowledge of its internal workings (Bakharev, 2023).

3. Methods

This chapter establishes the research methodology for investigating the causal impact of maternal healthcare interventions on maternal and neonatal health outcomes within Davao City from 2020 to 2024, accounting for geographic heterogeneity in treatment effects. The methodology integrates spatial analysis with modern causal inference techniques to identify how geographic factors dictate intervention effectiveness. It details the research design, data sources, data processing procedures, analytical techniques, dashboard output specifications, and validation strategies employed throughout the study.

3.1 Definition of Terms

This section defines key technical terms that will be utilized throughout this study. The following terms are as follows:

- Sparse or Unreliable Data — Datasets with less than 70% of its completeness or fewer than twelve valid monthly observations per indicator year are excluded from the analysis.

- Average Treatment Effect (ATE) — The estimated average causal impact of a maternal care intervention (e.g., facility-based delivery, prenatal visits) on the live birth rate across all barangays

- Conditional Average Treatment Effect (CATE) — The estimated treatment effect conditional on barangay-level covariates such as population density, facility access,

and health worker ratio, capturing variations in intervention performance across local contexts.

- Heterogeneous Treatment Effect (HTE) — Describes the variation in intervention effectiveness across barangays, identified through the Causal Forest model's variable importance metrics and Best Linear Projection (BLP) analysis.

- Falsification Test (Placebo Test) — A model validation process that applies the causal estimation method to outcomes theoretically unaffected by maternal interventions (e.g., adult male mortality rates) to confirm the absence of unobserved confounding.

- Spatial Cross-Validation — A validation approach that divides data into training and testing sets by geographic clusters (barangays or districts) rather than random sampling, to control for spatial autocorrelation and ensure reliable model performance.

- GeoJSON / Shapefiles — Digital file formats defining the geographic boundaries of barangays and districts, used for generating interactive heatmaps and spatial visualizations in the study's web dashboard.

- Causal Forest Algorithm — A Generalized Random Forest (GRF) method that estimates heterogeneous causal effects by partitioning covariate space to minimize treatment effect variance, enabling granular causal inference in spatial datasets.

- Best Linear Projection (BLP) — A post-modeling analysis that regresses estimated CATEs against covariates to determine which geographic and demographic features most significantly explain treatment effect variation.

- Imputation (Random Forest-Based) — A non-parametric technique applied to predict missing data values using relationships among existing variables, ensuring data completeness and consistency without discarding incomplete records.

- Routine Health Information System (RHIS) — The primary dataset provided by the Davao City Health Office (CHO), containing barangay-level maternal and child health data from 2020–2024, serving as the foundation for this study's analysis.

- Antenatal Intervention Variables — Quantitative indicators representing the proportion of total pregnancies that received a specific service (e.g., prenatal visits, iron supplementation), expressed as rates (e.g., rate_facility_delivery, rate_prof_attendant).

- Spatial Autocorrelation — The statistical tendency of neighboring barangays to exhibit similar maternal health outcomes, accounted for in this study through spatial cross-validation and geospatial modeling techniques.

- Live Birth Rate — The primary outcome variable, calculated as the ratio of live deliveries to total pregnancy outcomes per barangay, used to measure the causal effect of interventions.

- Geospatial Heterogeneity — The variation in intervention performance across barangays due to differences in geographic, infrastructural, and demographic contexts

3.2 Research Design

This study employed a quantitative, ecological research design aimed at assessing the causal relationship between maternal healthcare interventions and maternal and neonatal health outcomes across barangays in Davao City. The approach integrated spatial analysis with the Causal Forest algorithm, to investigate how geographic factors influence the effectiveness of interventions such as prenatal care, skilled birth attendance, and maternal supplementation.

This study utilized observational health records and historical data from routine monitoring data instead of randomized control trials. This study can leverage natural variation in intervention intensity across barangays. The study applied the causal inference techniques to estimate both Average and Conditional Average Treatment Effects (ATE and CATE). Geospatial data, health service records, and demographic features were all incorporated to reveal heterogeneity in outcomes.

This chapter outlines the research setting, data sources, data preparation process, model implementation, validation techniques, and the visualization system that translates findings into actionable insights.

The unit of analysis comprises aggregated data at the Barangay level. For indicators with sparse or unreliable data at the barangay level, the district level serves as the alternative unit of analysis.

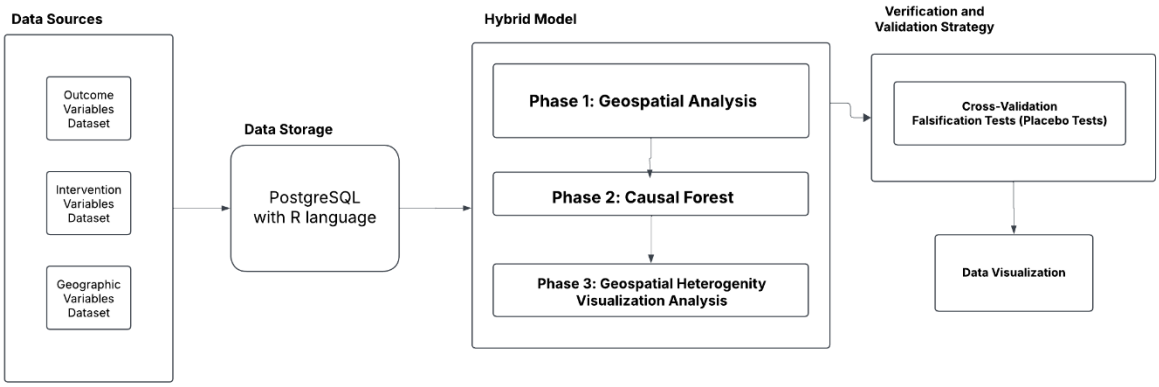

Figure 1: Conceptual Framework

Figure 1 illustrates the overall conceptual framework of this study. It outlines the sequential and interconnected stages, beginning with the acquisition of primary and secondary data sources. These data then undergo rigorous processing and management, leading into a multi-phase analytical approach designed to investigate spatio-temporal trends in maternal health. The outputs of this analysis, including identified patterns and statistical models, culminate in the development of an interactive dashboard. Crucially, the framework emphasizes that verification and validation strategies are applied throughout the entire research process to ensure the credibility and utility of the findings.

3.3 Research Setting and Time Frame

The research setting is Davao City, Davao Region, Philippines, encompassing the 182 barangays across 11 administrative districts. Due to availability of data, this study spans a five-year period from January 2020 to December 2024 utilizing historical data.

3.4 Population and Locale

The study population does not use a fixed sample size. Instead, it is defined by the administrative boundaries of Davao City during the study period, with particular focus on

women of reproductive age (15-49 years) and mothers/newborns represented in the maternal and child health (MCH) indicators. This encompasses all 182 barangays across the city's 11 districts.

The barangays of Davao City were selected as the primary unit of analysis as it represents the smallest administrative unit for which health interventions were systematically implemented, managed, and reported by the Davao City Health Office (CHO). This level of granularity was important for capturing the local heterogeneity in healthcare access, utilization, and outcomes that was central to the study's research questions. While household-level data offers greater detail, it was neither available for this city-wide study nor practical for a policy-level causal assessment, which raised significant data privacy concerns. Furthermore, an analysis at the municipal or district level would block the critical variations between individual barangays, thereby limiting the actionable specificity of our findings for targeted public health decision-making.

Figure 2: Davao City Districts and Barangays

3.5 Data Sources

This study utilized both primary routine health information system (RHIS) to ensure comprehensive analysis. The selection of multiple complementary data sources was essential to capture the multidimensional nature of maternal health outcomes and their determinants.

3.5.1 Primary Data: Davao City RHIS

This study uses the Davao City Routine Health Information System (RHIS) data. Although the RHIS is an administrative dataset, in this study it was considered as the primary dataset because it is the direct source collected from CHO before any cleaning or processing. This served as the foundation of this study for several critical reasons:

- It provides the most comprehensive, systematically collected data on maternal health services and outcomes in Davao City
- • It offers granular geographic detail (barangay-level)
- It contains longitudinal data allowing variation in exposures of maternal health interventions
- It represents the actual operational data used by health administrators for decision-making, ensuring research-to-practice relevance

The study relies on obtaining aggregated reports based on the standardized 2025 RHIS covering the period 2020-2024, obtained from the Davao City Health Office (CHO) through formal data sharing agreements.

3.5.2 Key Datasets

These are the key datasets that were identified as integral parts for the analysis. They contain the most relevant indicators to identify the disparities in maternal health outcomes.

- Maternal Care - Contains antenatal care, delivery, postpartum, complications, and adolescent pregnancy data. This template is essential as it captures the continuum of maternal care from pregnancy through delivery to postpartum, allowing for analysis of care continuity and drop-offs.
- Vital Statistics - Contains infant mortality rate, maternal mortality ratio, neonatal mortality rate, births, and deaths data. This data is crucial for outcome measurement and represents the gold standard indicators for maternal and child health globally.
- Demographics - Contains facility lists and health worker counts. This data enables analysis of healthcare access and human resource availability as key structural determinants.

3.5.3 Key MCH Indicators

The Maternal and Child Health indicators are the most important metrics to determine the effectiveness of maternal health care services.

- Antenatal Care Coverage Rate (ANC4+) - Selected as it predicts birth outcomes and reflects access to preventive care.
- Facility-Based Delivery Rate - Critical indicator for measuring institutional delivery access and utilization.
- Skilled Birth Attendant Rate - Gold standard global indicator for safe motherhood initiatives.
- Low Birth Weight Rate - Key outcome indicator reflecting both maternal nutrition and care quality.
- Adolescent Pregnancy Rate - Selected due to the elevated risk profile of this vulnerable population.
- Anemia Prevalence and Management Rate - Included as anemia is a leading contributor to maternal morbidity in the Philippines.

- Gestational Diabetes Mellitus (GDM) Prevalence and Management Rate - Selected due to rising prevalence and impact on outcomes.
- Stillbirth Rate - Critical outcome indicator often overlooked in traditional analyses

3.5.4 Secondary Data

Secondary data sources are essential to contextualize health indicators, calculate appropriate population-based rates, and examine social determinants of maternal health outcomes. The following datasets have been specifically selected to complement the primary RHIS data.

- Population Data: Official annual population estimates (2020-2024) per barangay, disaggregated by relevant age-sex groups from the Philippine Statistics Authority (PSA) and City Planning and Development Office (CPDO). These data are crucial for: Calculating accurate denominator values for population-based rates Standardizing indicators across areas with different population sizes Enabling age-specific analyses for targeted interventions Tracking population changes that might influence service utilization patterns
- Geospatial Data: Digital boundary maps (Shapefile/GeoJSON format) for Davao City barangays and districts from PSA, PhilGIS, and CPDO. Geospatial data is indispensable for: Enabling spatial statistical analyses including hotspot detection Creating meaningful geographic visualizations for stakeholders Calculating spatial accessibility measures such as distance to facilities Accounting for neighborhood effects and spatial spillovers in health outcomes Give context on the availability and accessibility of birthing and primary care facilities

3.6 Data Processing and Management

The study implements a systematic data processing workflow, as detailed in Figure 3, below, designed to transform raw administrative data into analysis-ready datasets while preserving data integrity and maximizing information extraction.

Figure 3: Data Processing and Management Plan

3.6.1 Data Acquisition & Consolidation

The nature of the data requires extensive selection and removal of irrelevant data points. The RHIS dataset is a large and comprehensive report on all public health related data. There is a large amount of data within the dataset that serves no purpose to the thesis as our focus is on maternal health.

The maternal health care dataset from RHIS is represented by nineteen Excel files for each of Davao City's health districts. While similar to administrative districts, health districts are further subdivided to ease service delivery and administration of public health. Within these files contains individual sheets for every month of the year and all the relevant maternal health statistics subdivided into individual barangays. With five years of data from 2020-2024, there is a total of ninety-five individual Excel files.

These files were created primarily to optimize data entry and readability. This results in structural issues that make it difficult to work with. Specifically, merged header cells, multiple header rows, embedded formatting, and references to external files. Steps need to be taken in order to manage these structural issues to ensure efficient data management.

There are 298 data rows in the excel file; however, most of these are subsets of other statistics that are divided by age-group or other subdivisions. There are also a significant number of rows that contain data that is irrelevant to the study. These will need to be managed accordingly so that only relevant data is collected.

The relevant data will be selected and transferred to a comprehensive database with all relevant data cleaned and ready for further analysis. These will then be compiled into more specific datasets to be implemented into causal forests.

A standardized script will be used to integrate the Excel files into R; the script includes importing the Excel files and managing the different sheets within them. Following that, the script will remove irrelevant formatting and fix column and row names.

This step is essential for integrating data from disparate sources with different structures. Creating structured integration protocols ensures consistent merging methodology across temporal and spatial dimensions. These protocols include standardized naming conventions, assigning unique identifiers to barangays/districts, and ensuring the handling of different levels of granularity.

3.6.2 Data Cleaning

The RHIS dataset is already a comprehensive and aggregated report that is submitted to the national government as such. However, because it is primarily formatted for reporting instead of analysis, extensive formatting adjustments are required.

However, due to RHIS being primarily a reporting system for decision makers, there is substantial formatting and redundant rows and columns for the sole purpose of readability and data entry. Before data cleaning when the Excel file is imported into R, there is a total of ~400 columns, most of these are just artifacts from the way it was formatted and presented in Excel.

These formatting artifacts will need to be carefully removed and excluded from the dataset. Column headers for each barangay are repeated multiple times within the dataset. There are multiple merged cells and entries that provide additional information to the data. Empty placeholder columns for each barangay that are used for data entry. The rows that exist solely to aggregate statistics that were divided by group such as age group, pregnancy trimester, or screened and diagnosed patients.

After removing all the formatting data and data entry columns, the resulting data set now only has 298 rows and the correct number of columns for barangays that exist within each district.

Following this initial structural cleaning, a second phase of analytical cleaning is performed. This involves:

- **Feature Selection:** Irrelevant columns (e.g., internal file trackers) and redundant variables are removed.
- **Preventing Data Leakage:** Variables that would leak information about the outcome are carefully excluded. For example, columns detailing the *type* of neonatal death are removed to avoid leaking information to the primary outcome variable.
- **Type Conversion:** All character-based columns, such as 'month', are converted into categorical factors to ensure they are correctly interpreted by the analytical models.
- **Identifier Separation:** High-cardinality identifiers like barangay and district are temporarily set aside during the imputation phase to prevent them from being used as predictors, only to be re-joined later.

3.6.3 Imputation of Missing Data

Routinely collected administrative data inevitably contains missing values. To address this without discarding valuable data (which could introduce bias), a machine-learning-based

imputation method is employed. This study utilizes a random forest-based imputation algorithm. This non-parametric method is highly effective as it can handle mixed data types (numerical and categorical) and capture complex, non-linear relationships within the data to make robust predictions for missing values.

3.6.4 Data Storage and Organization Framework

The aggregated dataset was stored to PostgreSQL, providing flexible data queries and seamless integration to the dashboard for data visualization. Data security and access control will remain under the discretion of the CHO.

3.7 Data Analysis

The analysis was conducted in three sequential phases using robust statistical and computational methods. Each analytical method had been carefully selected based on its suitability for addressing specific research questions, handling the characteristics of maternal health data, and specifically, for identifying heterogeneous treatment effects in a spatial context. The overarching goal is to transform processed data into actionable insights regarding intervention effectiveness across Davao City. Further details discussed the strategy of causal identification and the following phases to be conducted.

3.7.1 Causal Identification Strategy

The approach to this causal inference framework is based on the Rubin Causal Model, also known as the Potential Outcomes Framework, which defines the causal effect for a unit (e.g barangay) as the difference between its outcome under treatment and under control. The analysis aimed to answer the following causal question: “What is the causal effect t of

increasing the provision of a specific maternal intervention *W* (e.g, facility delivery rate) on the barangay-level live birth rate *Y,* for a barangay with a given set of characteristics *X*?".

As such, causal identification from observational data requires meeting key assumptions. The following conditions hold:

1. Conditional Unconfoundedness: A conditional on a rich set of observed barangay-level covariates X (including demographic profiles, healthcare workforce density, facility access, and temporal trends) which serves as the assignment of the treatment level W (intervention intensity) is independent of the potential outcomes, further assumes that it is measured and adjusted for all common causes of both the intervention uptake and the live birth rate.

2. Overlap: For every barangay, there is a positive probability of receiving any level of the treatment intensity, ensuring that there are comparable barangays across different levels of intervention implementation to estimate the treatment effect.

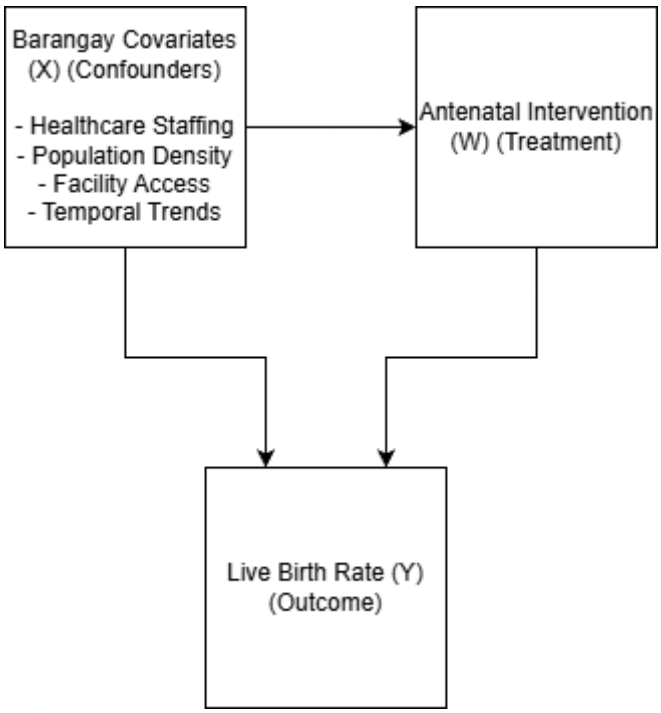

Figure 4: Directed Acyclic Graph

The graph above represents the concept of the causal structure to be conducted. The Causal Forest algorithm is well-suited to this context as it flexibly controls for this dimensional set of confounders X, making the unconfoundedness assumption more plausible than in traditional linear models.

Figure 4: Data Analysis Plan

3.7.2 Phase 1: Spatial Pattern Analysis

This initial phase focuses on characterizing the spatial distribution of key variables and identifying underlying geographic patterns that may influence both intervention implementation and health outcomes. This allowed deeper understanding of the research setting and informs the subsequent causal inference models.

Geographic Feature Engineering:

- This step involves creating new spatial variables that encapsulate critical aspects of access and infrastructure.
- Healthcare Worker-to-Population Ratios: For each barangay, calculate the ratio of different types of healthcare personnel (e.g., doctors, nurses, midwives, Barangay Health Workers) to its total population.

Facility Density and Distribution Metrics:

- Spatial Density: The number of birthing facilities per barangay. Categorization of Facilities: Facilities will be categorized and their distributions analyzed to understand the types of services available within different geographic areas.

- Spatial Clustering: This involves identifying areas with statistically significant concentrations of similar characteristics, which could indicate underlying spatial processes or disparities.

At the barangay level, choropleth maps, also known as heatmaps, were used to visualize the spatial distribution of all important outcome and intervention variables. The initial exploratory data analysis relied heavily on these visualizations, which made it possible to identify trends, apparent spatial patterns, and possible clusters of high and low values throughout Davao City. To ensure that geographic context was included in the causal inference framework, the feature set for the ensuing Causal Forest model was informed by the insights obtained from this descriptive spatial analysis as well as the engineered spatial covariates (such as healthcare worker density and facility access metrics).

3.7.3 Phase 2: Causal Forest Implementation

This phase is the core of causal inference, leveraging the strengths of Causal Forests to estimate the heterogeneous impacts of interventions. Causal Forests are particularly suited for this study due to their ability to (1) handle high-dimensional data with numerous covariates without requiring strong assumptions about their functional forms, (2) robustly estimate causal effects in the presence of complex confounding, and most importantly, (3) directly estimate heterogeneous treatment effects (HTE), allowing for the identification of how intervention effectiveness varies across different barangays based on their unique characteristics. This machine learning approach provides a non-parametric way to uncover nuanced causal relationships that traditional linear models might miss.

Model Specification:

To standardize all variables for the analysis, the raw counts for outcomes and interventions were converted into rates. This requires first establishing a consistent denominator for each unit of analysis (i.e., each barangay-month observation). This denominator, **total outcomes**, is calculated as the sum of all terminal pregnancy outcomes:

$$\boldsymbol{Total\ outcomes} = live\ deliveries + stillbirths + abortions$$

Outcomes (Y):

The primary outcome variable (Y) for the analysis is the live birth rate. This is calculated by dividing the count of live deliveries by the total outcomes denominator:

$$\boldsymbol{Y} = \frac{live\ deliveries}{total\ outcomes}$$

Treatments (W):

Similarly, all intervention variables (the "treatments") were converted from raw counts into rates. This represents the proportion of total pregnancies that received a specific service. Each intervention count was divided by the same total outcomes denominator, ensuring a consistent scale.

For example, the rate for receiving at least 4 prenatal visits is calculated as:

$$\frac{\boldsymbol{W} = deliveries\ with\ 4\ prenatal\ visits}{total\ outcomes}$$

This method is applied to all 21 intervention variables, which are grouped as follows:

- **Prenatal Care & Planning:**
    - Pregnant women with first prenatal visit
    - Pregnant women with at least 4 prenatal visits
    - Pregnant women given birth plan
    - Pregnant women given iodine

- Pregnant women given deworming tablets
- Pregnant women given counseling
- Pregnant women seen by a doctor

- **Supplements:**
  - Pregnant women who completed calcium carbonate supplements
  - Pregnant women with complete iron with folic acid supplements
  - Pregnant anemic women given iron supplements
- **Screening & Lab Tests:**
  - Pregnant women screened for syphilis
  - Pregnant women tested for hepatitis B
  - Pregnant women tested for HIV
  - Pregnant women tested for complete blood count
  - Pregnant women screened for gestational diabetes
  - Pregnant women underwent hemoglobin determination
  - Pregnant women with blood typing
  - Pregnant women underwent urinalisys
  - Pregnant women with 3 basic laboratory exams

Attendant & Locations is calculated with slightly different formulas

- **Attendant & Location:**
  - Facility based delivery

$$W = \frac{deliveries\ in\ government\ facility\ +\ deliveries\ in\ private\ facility}{total\ outcomes}$$

  - Trained birth attendant present during delivery

$$W = \frac{Birth\ attended\ by\ doctor\ +\ birth\ attended\ by\ nurse\ +\ birth\ attended\ by\ midwife}{total\ outcomes}$$

Each of these 21 intervention rates is analyzed individually as a "treatment" variable (W) against the live birth rate (Y) to estimate its causal effect.

Covariates (X):

- Demographic characteristics: Age distribution of birthing mothers, barangay population density, total barangay population.
- Facility access and geographic features: The engineered variables from Phase 1, including healthcare worker-to-population ratios, facility density, and barangay spatial categories. These are crucial for understanding geographic heterogeneity.
- Temporal indicators: Year and month fixed effects to account for time trends or seasonality in health data.

These were the potential confounders and effect modifiers that influence both intervention uptake and health outcomes and are critical for achieving causal identification.

Causal Forest Training:

The Causal Forest model was trained using the grf package in R (or econml.grf in Python), which implemented Generalized Random Forests. This involved growing an ensemble of causal trees, where each tree was built by recursively partitioning the covariate space to minimize the variance of treatment effect estimates within each leaf node.

Hyperparameter Tuning:

Key hyperparameters such as the number of trees (num.trees), minimum node size (min.node.size), and subsampling rates (sample.fraction) were tuned using cross-validation to optimize model performance and ensure robust estimation.

Treatment Effect Estimation:

- Average Treatment Effects (ATE): For each intervention, the overall average causal impact on the various maternal and neonatal health outcomes across all barangays will be estimated. This provides a general understanding of the intervention's efficacy.
- Conditional Average Treatment Effects (CATE): This is the primary output of interest for addressing the study's core question on heterogeneity. Causal Forests will estimate the CATE for each intervention at the individual barangay level, conditional on its unique profile of covariates (including all geographic features). This allows for a granular understanding of how the impact of a specific intervention varies from one barangay to another.
- Heterogeneous Treatment Effects across Geographic Features: The estimated CATEs will then be analyzed to explicitly identify how the effectiveness of interventions differs based on specific geographic features This analysis will highlight which geographic contexts are most receptive or resistant to interventions.

3.7.4 Phase 3: Geospatial Heterogeneity Visualization Analysis

This delves into understanding the drivers of the heterogeneous treatment effects identified in the Causal Forest Model, focusing heavily on geographic factors.

Variable Importance Measures:

- Quantifying Predictive Power: The importance of each geographic feature in explaining the variation in estimated treatment effects will be formally assessed. This will be done by examining how much each covariate contributes to the heterogeneity in the estimated CATEs within the Causal Forest framework (e.g., using "feature importance" metrics specifically designed for HTE estimation in causal forests, which identify variables that predict differences in treatment effects).

- Comparison of Spatial vs. Non-Spatial Variables: A direct comparison will be made to ascertain whether spatial variables are more predictive of treatment effect variation than traditional demographic or socio-economic variables. This will highlight the unique contribution of the spatial dimension to understanding intervention effectiveness.

Interaction Analysis:

- Identifying Thresholds and Synergies: While causal forests implicitly account for complex interactions, this phase will explicitly analyze key interactions to provide actionable insights. This includes:
- Role of Facility Type and Personnel Allocation: Investigating if specific types of facilities or higher allocations of particular healthcare personnel amplify or diminish the effects of certain interventions, especially in remote or underserved areas.

3.8 Verification and Validation Strategy

Rigorous verification and validation strategies were employed throughout the analysis to ensure the robustness, reliability, and generalizability of the study's findings, particularly concerning the causal claims in a spatial context.

Falsification Tests (Placebo Tests):

- Applying Methodology to Unaffected Outcomes: The causal forest methodology was applied to outcomes that were theoretically unlikely to be affected by the maternal healthcare interventions (e.g., adult male mortality rates, incidence of non-maternal/neonatal specific communicable diseases that do not directly involve the maternal health system). If significant causal effects are detected for these "placebo"

outcomes, it can indicate potential unobserved confounding or issues with the causal identification strategy.

- Testing for Effects Before Intervention Implementation: Where historical data allows, the methodology was applied to data from a period before the substantial implementation of the interventions being studied. The absence of significant "treatment effects" during this pre-intervention period would provide stronger evidence that the observed effects are genuinely attributable to the interventions rather than pre-existing trends or unobserved factors.

3.9 Robustness and Validation Strategy

To ensure the reliability of causal estimates and to rule out false correlations, this study employs a validation strategy that further tests the key assumptions of the model.

1. Falsification Tests (Placebo Test)

Two types of falsification tests will be conducted to determine potential unobserved confounding.

a. Placebo Treatment Test: The real intervention variable W mentioned in 3.7.1 will be replaced with a randomly generated, placebo treatment. A valid model should correctly estimate with a null effect (ATE ≈ 0) for this fake intervention.

b. Placebo Outcome Test: The model is applied to an outcome variable that is theoretically unrelated to maternal interventions, specifically the *projected_population* of the barangay. Finding an important effect would indicate the model is detecting patterns driven by underlying confounders rather than true causal relationships.

2. Spatial Cross-Validation

To prevent overbearing performance metrics due to spatial autocorrelation (e.g nearby barangays have similar characteristics), the data is partitioned for model tuning and evaluation using a spatial blocking strategy. Rather than random train-test splits, it is held out entire districts during training, ensuring the model's performance is evaluated on geographically distinct areas, providing a more realistic measure of its generalizability and robustness.

3. Best Linear Projection (BLP) Analysis

After estimate the Conditional Average Treatment Effects (CATEs), BLP is used to regress these individual treatment effects against the baseline covariates X, in which this analysis serves as a key diagnostic tool where:

a. It tests whether the heterogeneity discovered by the Causal Forest is meaningfully associated with observable barangay characteristics (e.g., does a higher nurse count predict a larger treatment effect?).

b. It provides an interpretable summary of how and why the treatment effect varies, adding a layer of face validity to the model's complex output.

3.10 Ethical Considerations

The study adheres to the highest ethical standards in health research, prioritizing data privacy, confidentiality, and responsible dissemination of findings.

Data Aggregation and De-identification: The study will exclusively use aggregated, de-identified data at the barangay level. No individual patient information will be accessed, processed, or stored at any point. This ensures that the privacy of individuals is fully protected.

Formal Data Sharing Agreements: A formal, Data Sharing Agreement (DSA) will be established with the Davao City Health Office (CHO). This agreement will explicitly outline

the terms of data access, usage, storage, retention, and destruction, ensuring compliance with the CHO's data governance policies and national data privacy laws.

Responsible Reporting and Dissemination: Research findings will be reported responsibly, focusing on public health implications and avoiding any form of stigmatization of specific geographic areas or populations. Results will be presented in a way that is culturally sensitive and promotes equitable health interventions. Complete findings, including the developed dashboard, will be shared proactively with the Davao City Health Office and other relevant public health stakeholders to inform evidence-based public health action and policy development.

3.11 Software and Tools

The analysis will primarily utilize powerful open-source programming languages and specialized libraries, complemented by Geographic Information System (GIS) software and robust data management tools.

Programming Languages:

Python 3.9+:

- Data Manipulation: Pandas 1.4.0+ for efficient data cleaning, transformation, and aggregation.
- Geospatial Analysis: GeoPandas 0.10.0+ for working with spatial dataframes, performing spatial joins, and manipulating geographic features.
- Statistical Modeling: Statsmodels 0.13.0+ for complementary statistical tests and basic regression analysis.
- Visualization: Matplotlib, Seaborn, and Plotly for creating static and interactive data visualizations to present the analysis results.

- API Routing: Flask framework and Cross-Origin Resource Sharing (CORS) to route requests to and from the frontend framework.

R 4.1.0+:

- Data Processing: tidyverse suite (e.g., dplyr, ggplot2, tidyr) for efficient data manipulation, visualization, and consistent syntax.
- Causal Inference: grf (Generalized Random Forests) package for robust implementation of Causal Forests and other causal machine learning methods.

GIS Software:

- QGIS 3.22+: A free and open-source desktop GIS application for advanced geospatial processing, digitizing, geocoding health facility addresses, creating detailed basemaps, performing network analysis for distance calculations and cartographic production for thesis figures.

Database:

- PostgreSQL: This will serve as the primary database to hold the aggregated dataset to fetch data to the dashboard.

3.12 Data Visualization

Figure 5: Davao City Barangays and Districts Shapefiles exported via GeoJSON from QGIS

Like other related studies, the best way to comprehend data is visualization through heatmaps by creating a dashboard. The dashboard consists of a map panel that only shows Davao City as well as its surrounding districts and barangays via GeoJSON shapefiles exported from QGIS, as well as card panels to display number statistics of the dataset. The dataset, “philippines-psgc-shapefiles" by *altcoder* found on Github, is used for the shapefiles to trace out barangays found in Davao City, whereas OpenStreetMaps via overpass-turbo is used for tracing districts. This method saves performance on the app as the GeoJSON files are saved locally, rather than relying on APIs to fetch maps and shapefiles. In an administrative level to classify a country, barangays are classified as level 4, whereas districts are classified as level 2.

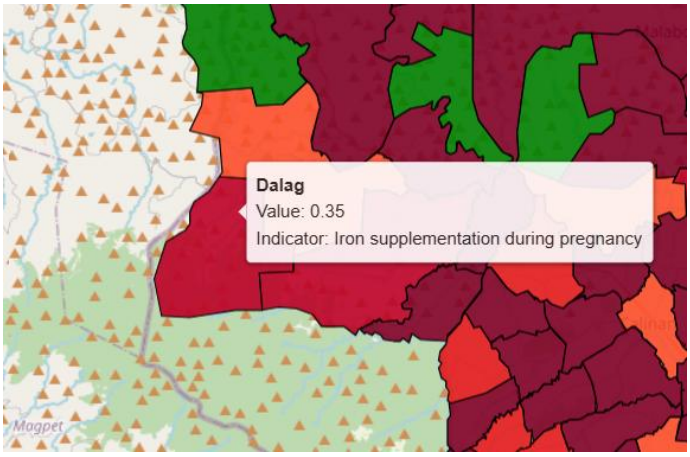

Figure 6: Sample Barangay Heatmap with Top Indicator and Value

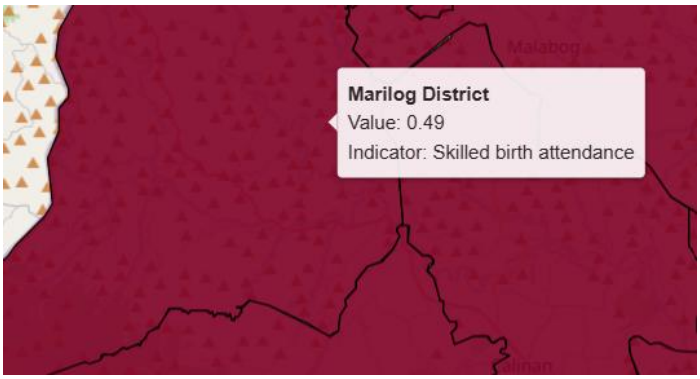

Figure 7: Sample District Heatmap with Top Indicator and Value

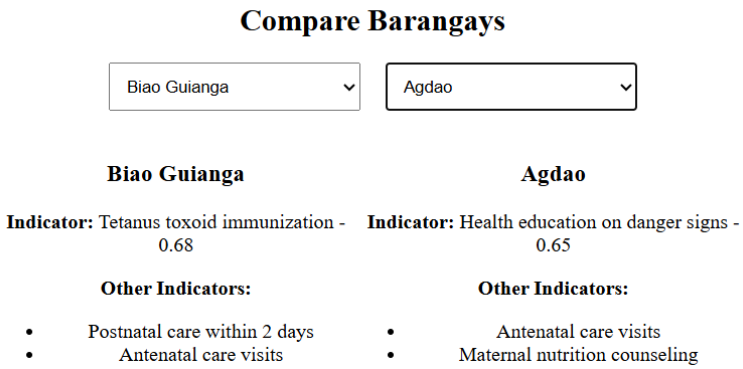

Figure 8: Comparing Barangay A and B

The frontend will be developed using JavaScript with the React framework under Vite, providing a lightweight and fast development environment. The backend will be powered by Python using the Flask framework, which is responsible for serving the GeoJSON shapefiles,

handling API requests to fetch and process relevant data (e.g., maternal care indicators, barangay and district names, numerical data) and building the proposed model. These data values are bound to geographic units and visualized as color gradients on the map using heatmaps, allowing users to visually identify areas of concern or interest. Interactive features such as showing either barangays or districts on the map panel, as well as a component for comparing barangay A and B, will be implemented to enhance user engagement and support decision-making. Cards displayed on the right side of the map panel will provide an overview of the dataset, including numbers of total morbidity and mortality cases, along with other relevant numerical indicators. The data presented in the figures above are for illustrative purposes only and do not represent the actual dataset to be used in the study.

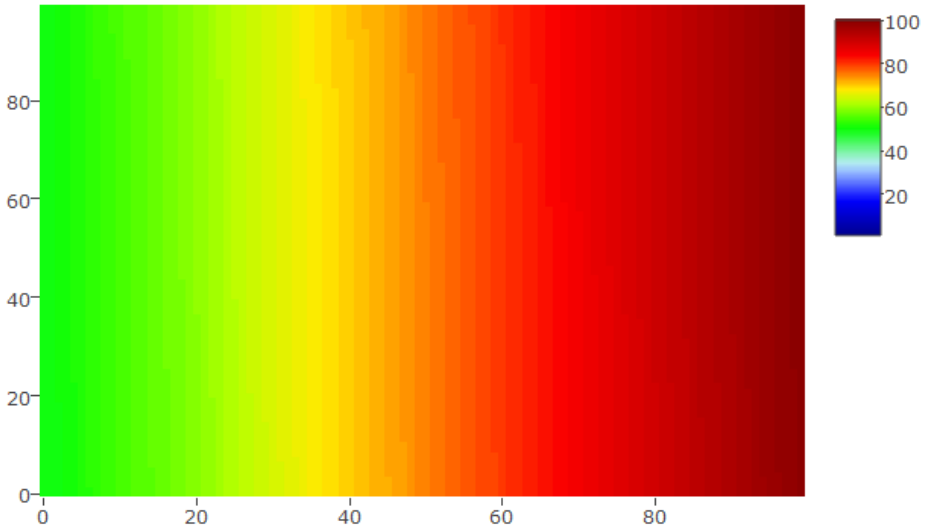

Figure 9: Sample Green-to-Red Heatmap

The heatmaps will indicate the severity of predicted indicators based on the trained model. Each barangay will display the value of its top indicator, as well as other indicators based on its hierarchical value in descending order, and be classified as Safe (green), Caution (yellow), or Act (red).

4. Results and Discussion

This chapter presents the quantitative findings of the Causal Forest analysis. The primary objective is to estimate the causal impact of various antenatal interventions on the barangay-level live birth rate (rate_live_birth). The chapter is organized into five sections:

1. **Key Variable Definitions:** A reference for the intervention variables used in the analysis.
2. **Average Treatment Effects (ATE):** The overall, average causal impact of each intervention across all of Davao City.
3. **Heterogeneous Treatment Effect (HTE) Analysis:** An exploration of *why* and *where* the effects of the top-performing interventions vary, based on covariate importance and a Best Linear Projection (BLP) against geographic and temporal variables.
4. **Model Validation:** A summary of falsification tests conducted to ensure the model's robustness and validity.
5. **Dashboard Visualization:** Given results visualized on a heatmap with comparisons between each district and barangay.
6. **Discussion of Findings:** A high-level interpretation of the results and their implications.

4.1 Key Variable Definitions

The analysis in this chapter measures the causal effect of 21 distinct antenatal interventions. These interventions, referred to by their variable names from the analysis script, are defined in Table 2.

Table 2: *Definitions of Antenatal Intervention Variables*

Variable Name	Description
rate_facility_delivery	Rate of deliveries that occurred in either a government or private health facility.
rate_prof_attendant	Rate of births attended by a health professional (doctor, nurse, or midwife).
rate_health_counseling	Rate of deliveries where the mother received health information and counseling.
rate_4_prenatal_visits	Rate of deliveries where the mother completed at least four prenatal visits.
rate_lab_test_basic_three	Rate of pregnant women who completed three basic lab tests (Hemoglobin, Urinalysis, Syphilis screen).
rate_completed_iron_folic	Rate of deliveries where the mother completed the iron with folic acid supplement regimen.
rate_lab_test_blood_typing	Rate of pregnant women who received a blood typing test.
rate_visit_1st_tri	Rate of deliveries where the mother's first prenatal visit was in the first trimester.
rate_lab_test_hemoglobin	Rate of pregnant women who were tested for hemoglobin (standalone test).
rate_lab_test_urinalysis	Rate of pregnant women who completed a urinalysis test.
rate_seen_by_doctor_preg	Rate of deliveries where the mother was seen by a doctor during pregnancy.
rate_completed_calcium	Rate of deliveries where the mother completed calcium supplements.
rate_tested_cbc_hgb	Rate of pregnant women who were tested for CBC/Hgb.
rate_birth_plan	Rate of pregnant women who created a birth plan.
rate_screened_gdm	Rate of pregnant women who were screened for gestational diabetes.
rate_given_deworming	Rate of pregnant women who were given deworming.
rate_screened_syphilis	Rate of pregnant women who were screened for syphilis.
rate_tested_hep_b	Rate of pregnant women who were tested for Hepatitis B.
rate_anemic_given_iron	Rate of anemic pregnant women who were given iron supplements.
rate_tested_hiv	Rate of pregnant women who were tested for HIV.
rate_given_iodine	Rate of pregnant women who were given iodine supplements.

4.2 Average Treatment Effects (ATE)

The Average Treatment Effect (ATE) represents the estimated average causal increase in the live birth rate if a barangay were to increase its application of a specific intervention. The Causal Forest model was run independently for each of the 21 antenatal interventions.

The complete ATE results are presented in **Table 2**, sorted by the ATE point estimate in descending order. The ATE column shows the causal effect, while the 95% confidence interval indicates the range of plausible values. An intervention is considered statistically significant if its confidence interval does not cross zero.

Table 3: *Causal Forest ATE for All Antenatal Interventions on Live Birth Rate*

Intervention	ATE	Std. Error	Lower 95% CI	Upper 95% CI	Significant?
rate_facility_delivery	**0.0449**	**0.0214**	**0.0029**	**0.0868**	**Yes**
rate_prof_attendant	**0.0445**	**0.0232**	**-0.0009**	**0.0900**	**No***
rate_health_counseling	**0.0286**	**0.0171**	**-0.0049**	**0.0622**	**No**
rate_4_prenatal_visits	**0.0253**	**0.0094**	**0.0069**	**0.0436**	**Yes**
rate_lab_test_basic_three	**0.0238**	**0.0123**	**-0.0002**	**0.0479**	**No***
rate_visit_1st_tri	**0.0207**	**0.0095**	**0.0021**	**0.0394**	**Yes**
rate_lab_test_hemoglobin	**0.0202**	**0.0085**	**0.0036**	**0.0368**	**Yes**
rate_lab_test_urinalysis	**0.0196**	**0.0076**	**0.0046**	**0.0346**	**Yes**
rate_seen_by_doctor_preg	**0.0191**	**0.0065**	**0.0063**	**0.0318**	**Yes**
rate_completed_iron_folic	**0.0189**	**0.0096**	**0.0000**	**0.0377**	**No***
rate_given_iodine	**0.0186**	**0.0125**	**-0.0059**	**0.0431**	**No**
rate_lab_test_blood_typing	**0.0171**	**0.0142**	**-0.0107**	**0.0450**	**No**

rate_screened_syphilis	**0.0163**	**0.0135**	**-0.0101**	**0.0426**	**No**
rate_birth_plan	**0.0123**	**0.0141**	**-0.0154**	**0.0400**	**No**
rate_tested_cbc_hgb	**0.0115**	**0.0190**	**-0.0258**	**0.0488**	**No**
rate_screened_gdm	**0.0081**	**0.0098**	**-0.0112**	**0.0274**	**No**
rate_completed_calcium	**0.0062**	**0.0070**	**-0.0075**	**0.0200**	**NYo**
rate_given_deworming	**0.0045**	**0.0110**	**-0.0171**	**0.0261**	**No**
rate_anemic_given_iron	**0.0034**	**0.0121**	**-0.0203**	**0.0271**	**No**
rate_tested_hep_b	**0.0021**	**0.0132**	**-0.0237**	**0.0279**	**No**
rate_tested_hiv	**-0.0017**	**0.0158**	**-0.0326**	**0.0293**	**No**

Note: Interventions marked No (e.g., rate_prof_attendant, rate_lab_test_basic_three, rate_completed_iron_folic) have confidence intervals that just barely include zero, suggesting a potential positive effect that does not meet the strict $p < 0.05$ threshold for statistical significance in this model.*

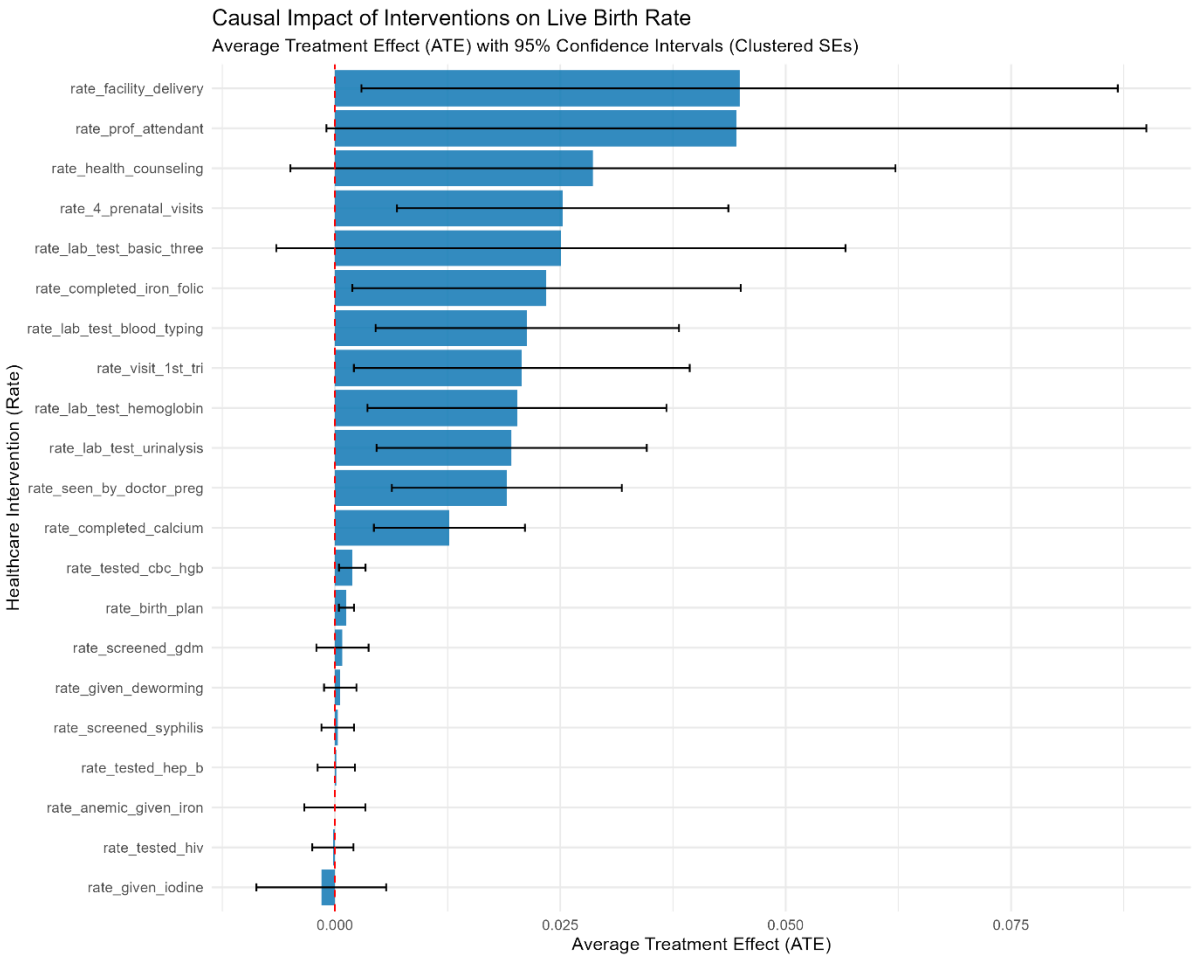

Figure 10: A visual overview of these average treatment effects

The results indicate that rate_facility_delivery (ATE: 0.0449) is the most impactful intervention with statistical significance, suggesting that an increase in facility-based deliveries is causally linked to the highest average increase in the live birth rate.

Following this are several other interventions that also show a significant positive effect:

- rate_4_prenatal_visits (ATE: 0.0253)
- rate_visit_1st_tri (ATE: 0.0207)
- rate_lab_test_hemoglobin (ATE: 0.0202)
- rate_lab_test_urinalysis (ATE: 0.0196)
- rate_seen_by_doctor_preg (ATE: 0.0191)

Interestingly, rate_prof_attendant (ATE: 0.0445) shows an effect size nearly identical to rate_facility_delivery, but its wider confidence interval (-0.0009 to 0.0900) prevents it from being classified as statistically significant. This finding is explored further in the discussion section. The remaining 14 interventions show positive point estimates that are smaller and not statistically significant.

4.3 Heterogeneous Treatment Effect (HTE) Analysis

While the ATE provides a useful city-wide average, the Causal Forest model is primarily designed to identify *heterogeneity* in treatment effects. This analysis explores how the causal impact of an intervention changes based on the specific characteristics of a barangay.

This section focuses on the three interventions with the largest positive ATE point estimates:

1. rate_facility_delivery
2. rate_prof_attendant
3. rate_4_prenatal_visits

For each intervention, we first identify the most important covariates in predicting heterogeneity (Variable Importance) and then analyze how the effect varies across geographic districts and time (Best Linear Projection).

4.3.1 HTE Analysis: rate_facility_delivery

Variable importance measures which covariates are the strongest predictors of *variation* in the treatment effect. **Table 4** and **Figure 10** show the top covariates for rate_facility_delivery.

Table 4: *Top Covariates Predicting HTE for rate_facility_delivery*

Rank	Covariate	Importance Score

1	num_nurses	0.0931
2	num_midwives	0.0917
3	num_doctors	0.0787
4	num_med_techs	0.0762
5	projected_population	0.0706
6	num_dentists	0.0685
7	num_nutritionists	0.0634
8	num_barangays	0.0488

The analysis reveals that barangay-level healthcare staffing—specifically the number of nurses, midwives, and doctors—are the most critical factors in explaining the variation in the causal effect of facility deliveries.

Best Linear Projection (BLP)

The BLP analyzes the relationship between the estimated treatment effect and specific covariates of interest (district, month, year). This identifies *where* and *when* the intervention is more or less effective. **Table 5** shows the statistically significant coefficients from the BLP.

Table 5: *Significant BLP Coefficients for rate_facility_delivery*

Covariate	**Estimate**	**Std. Error**	**p-value**
(Intercept)	0.2319	0.0682	0.00067
district_BUHANGIN	-0.2307	0.0886	0.00924

district_CALINAN	0.3207	0.1068	0.00267
district_PAQUIBATO	-0.3228	0.1287	0.01202
district_TUGBOK	-0.2227	0.1086	0.04013
month_AUGUST	-0.1501	0.0734	0.04077

The results show significant geographic heterogeneity. The baseline effect (Intercept) is positive and strong. However, relative to the baseline (Agliano District), the effect of facility delivery is significantly *lower* in the Buhangin, Paquibato, and Tugbok districts. Conversely, the effect is significantly *higher* in the Calinan district. A significant temporal effect is also noted, with the intervention's impact being lower in August.

4.3.2 HTE Analysis: rate_prof_attendant

Variable Importance

Figure 11 and Table 6 show the top **covariates for** rate_prof_attendant.

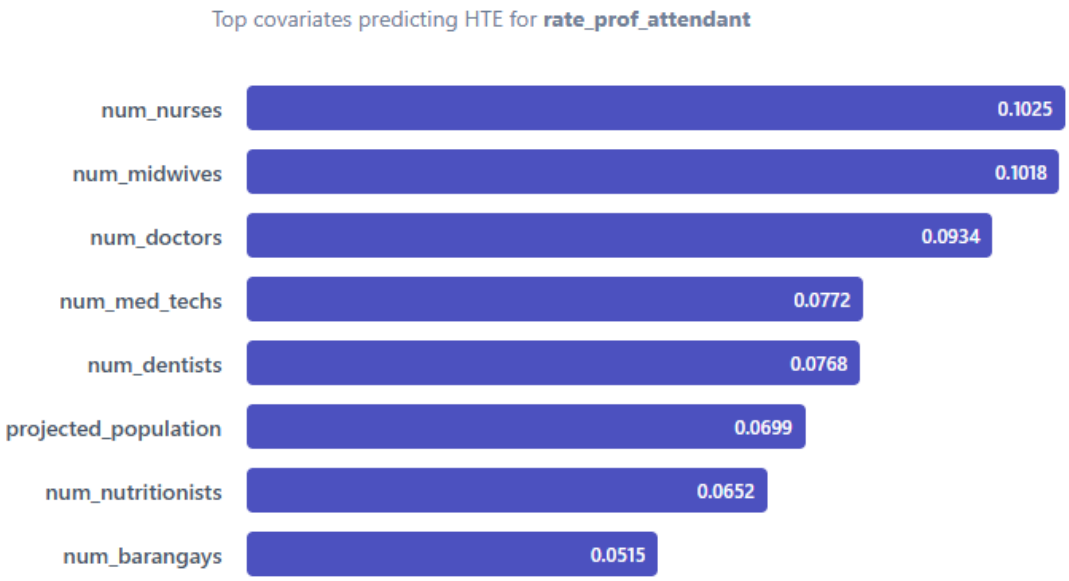

Figure 11: Variable Importance for rate_prof_attendant

Table 6: *Top Covariates Predicting HTE for rate_prof_attendant*

Rank	Covariate	Importance Score
1	num_nurses	0.1025
2	num_midwives	0.1018
3	num_doctors	0.0934
4	num_med_techs	0.0772
5	num_dentists	0.0768
6	projected_population	0.0699
7	num_nutritionists	0.0652
8	num_barangays	0.0515

The findings are highly consistent with the previous intervention. Healthcare staffing levels (num_nurses, num_midwives, num_doctors) are again the most dominant predictors of heterogeneity.

Best Linear Projection (BLP)

Table 7 shows the statistically significant BLP coefficients for rate_prof_attendant.

Table 7: *Significant BLP Coefficients for rate_prof_attendant*

Covariate	Estimate	Std. Error	p-value
(Intercept)	0.1983	0.0717	0.00561
district_TALOMO_SOUTH	-0.1740	0.0811	0.03194

The geographic heterogeneity for this intervention is less pronounced than for facility delivery. However, a significant negative coefficient is found for district_TALOMO_SOUTH, indicating the causal effect of having a professional attendant is weaker in this district compared to the baseline.

4.3.3 HTE Analysis: rate_4_prenatal_visits

Variable Importance

Table 8 and **Figure 12** show the top covariates for rate_4_prenatal_visits.

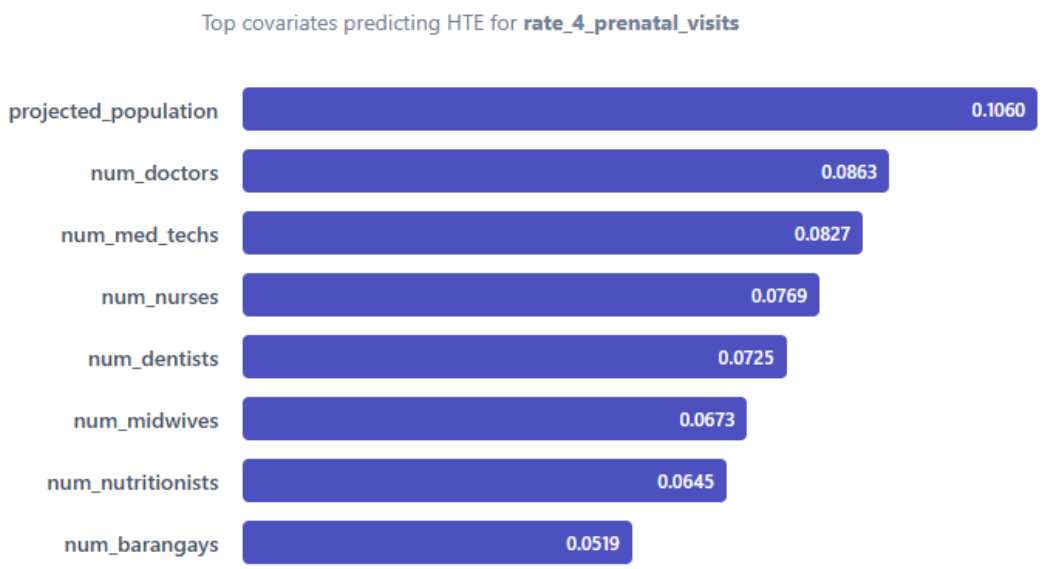

Figure 12: Variable Importance rate_4_prenatal_visits

Table 8: *Top Covariates Predicting HTE for rate_4_prenatal_visits*

Rank	Covariate	Importance Score
1	projected_population	0.1060
2	num_doctors	0.0863
3	num_med_techs	0.0827
4	num_nurses	0.0769
5	num_dentists	0.0725
6	num_midwives	0.0673
7	num_nutritionists	0.0645
8	num_barangays	0.0519

For this intervention, projected_population emerges as the single most important predictor of heterogeneity, followed by the familiar pattern of healthcare staffing levels.

Best Linear Projection (BLP)

Table 9 shows the statistically significant BLP coefficients for rate_4_prenatal_visits.

Table 9: *Significant BLP Coefficients for rate_4_prenatal_visits*

Covariate	Estimate	Std. Error	p-value
district_BAGUIO	0.1322	0.0573	0.0210
month_FEBRUARY	0.0739	0.0698	0.2894
month_JULY	0.1634	0.0864	0.0585

The BLP results show a significant positive effect for district_BAGUIO, indicating the intervention is *more* effective in this district. A positive temporal trend is suggested for July, though it falls just outside the $p < 0.05$ threshold.

4.4 Model Validation: Falsification Tests

To ensure the Causal Forest model is robust and genuinely identifying causal relationships rather than mere correlations or model artifacts, two falsification tests were performed for each of the top three interventions, as recommended in causal inference literature. The results are summarized in **Table 10**.

Table 10: *Falsification Test Results for Top 3 Interventions*

Intervention	Test	ATE	Std. Error	p-value	Result
rate_facility_delivery	Placebo Treatment	0.0051	0.0213	0.814	**Passed**

	Placebo Outcome	-0.0016	0.0102	0.873	**Passed**
rate_prof_attendant	Placebo Treatment	-0.0163	0.0733	0.822	**Passed**
	Placebo Outcome	-0.0022	0.0093	0.804	**Passed**
rate_4_prenatal_visits	Placebo Treatment	-0.0052	0.0345	0.879	**Passed**
	Placebo Outcome	0.0076	0.0354	0.827	**Passed**

1. **Placebo Treatment Test:** This test assesses whether the model would "find" a causal effect for an intervention that is known to be random noise. For all three top interventions, the resulting ATE was near-zero (e.g., -0.0051, -0.0163, -0.0052) with high p-values (all > 0.81). This confirms that the model does not identify a causal effect where one does not exist.
2. **Placebo Outcome Test:** This test assesses whether the real intervention appears to have a causal effect on an outcome it should not be related to (projected_population). Again, for all three interventions, the ATE was near-zero and highly non-significant (all p-values > 0.80). This confirms that the model is correctly isolating the effect to its specific, relevant outcome.

All six falsification tests passed decisively, providing strong support for the validity of the model and the causal nature of the findings presented in Section 4.2 and 4.3.

4.5 Dashboard Visualization

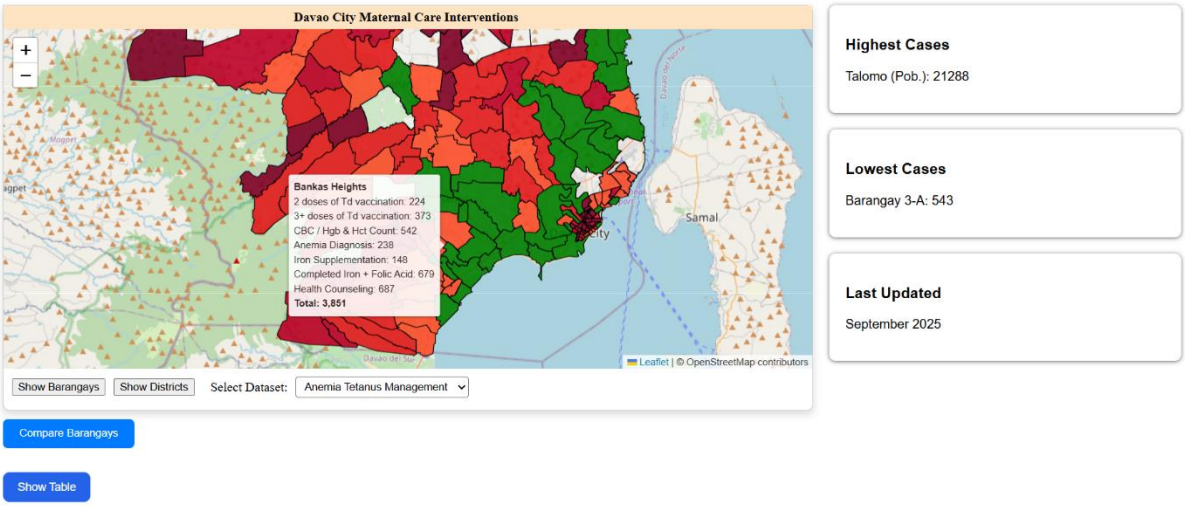

Figure 13: Overall Dashboard

The dashboard fetches data primarily from the Flask framework, which is built to gather map data via GeoJSON and numerical data from the Causal Forest via PostgreSQL. On the front-end, it is built using React under Vite, which focuses on building separate components to easily combine in one webpage. Because of this, React was chosen as a basis for the front-end for its efficiency and performance as well as its component-based architecture; it is easier to debug code along the way. Furthermore, Flask is used on the back end not just for its performance, but also for its compatibility with Python. The libraries found in Python make it easier to function with R, which makes the implementation of the software more fluid. The panels that will be discussed below are each placed individually as components upon implementation of the software.

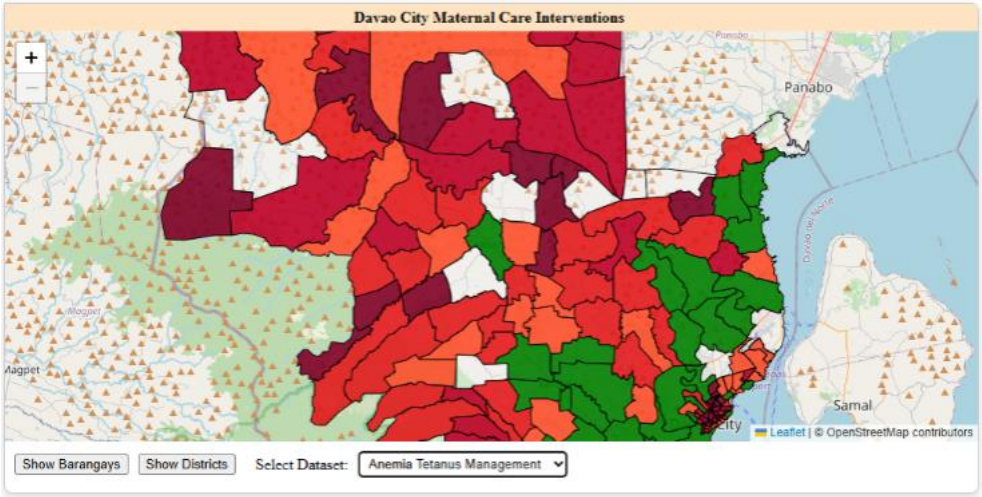

Figure 14: Map panel

The map panel consists of 182 barangays and 11 districts found in Davao City. It is using the leaflet library to generate the map itself. By using GeoJSON, it can trace edges per barangay and district that will further distinguish between one another. Every barangay and district has distinct values and descriptions based on the given aggregated dataset. Not all barangays are highlighted however as these barangays are not included in the dataset, either due to nonexistent data or their barangays have no health facilities whatsoever. Despite this, every barangay is listed and grouped into districts in which all districts are then highlighted.

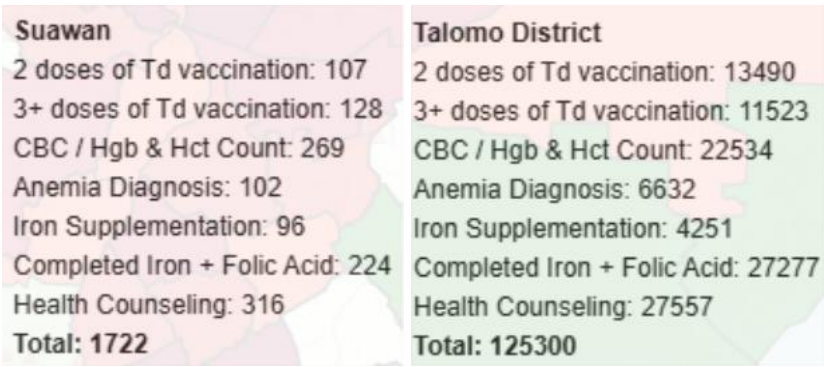

Figure 15: Sample data per barangay displayed when hovered by mouse cursor

Upon hovering the mouse cursor to each barangay and district displays the data based on a given dataset. The values given are summed up from 2020 to 2024.

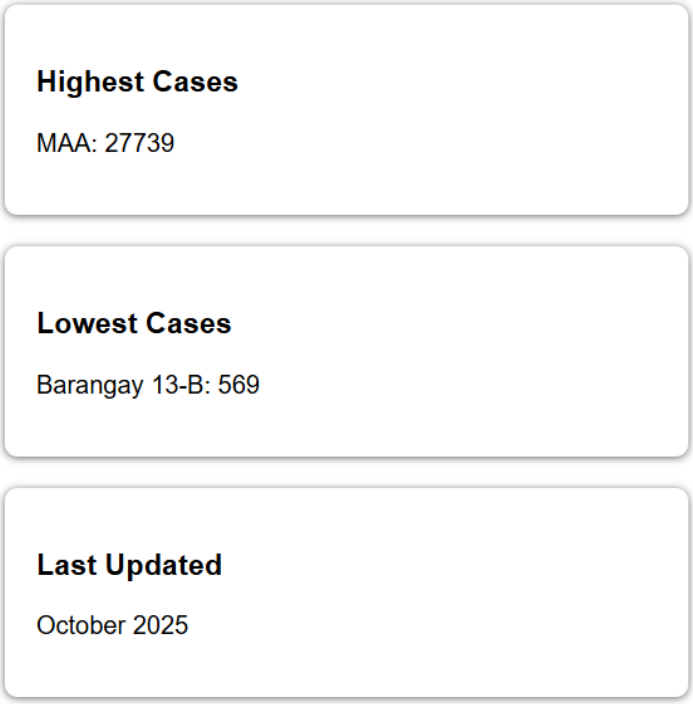

Figure 16: General Statistic displayed in panels

The panels found on the right side are general statistics of the dataset, from the number of cases to the highest-ranking intervention found. Based on the highlighted dataset in the map panel, it dynamically changes the total number of cases found in the dataset.

Anemia Tetanus Management | Barangays | Agdao

Anemia Tetanus Management for Agdao

Barangay	Year	2 doses of Td vaccination	3+ doses of Td vaccination	CBC / Hgb & Hct Count	Anemia Diagnosis	Iron Supplementation	Completed Iron + Folic Acid	Health Counseling
Barangay AGDAO	2020	196	167	192	2	1	164	170
	2021	229	110	191	25	0	235	256
	2022	424	314	615	80	80	398	398
	2023	161	88	306	26	26	231	220
	2024	45	30	63	35	35	88	114
	Total Years	**1055**	**709**	**1367**	**168**	**142**	**1116**	**1158**
Grand Total (All Barangays, All Years)		**1055**	**709**	**1367**	**168**	**142**	**1116**	**1158**

Figure 17: Detailed dataset in table displays

The table displays the dataset in detail. It showcases data that are based on the datasets mentioned in Chapter 3.

The software is provided with datasets that cover results based on the given model. Each dataset represents a different scenario ranging from Post Partum Care and Natality to Birth Facilities and Health Districts. In total, there are 9 datasets utilized in this study. The results that will be showcased are based primarily on the barangay level (except “Health Districts”) as it is the focus on this study. Furthermore, the heatmap is comprehended differently from its highest or lowest total value. Each dataset to be discussed will only show the highest and lowest total number of cases among all barangays in Davao City.

4.5.1 Anemia Tetanus Management

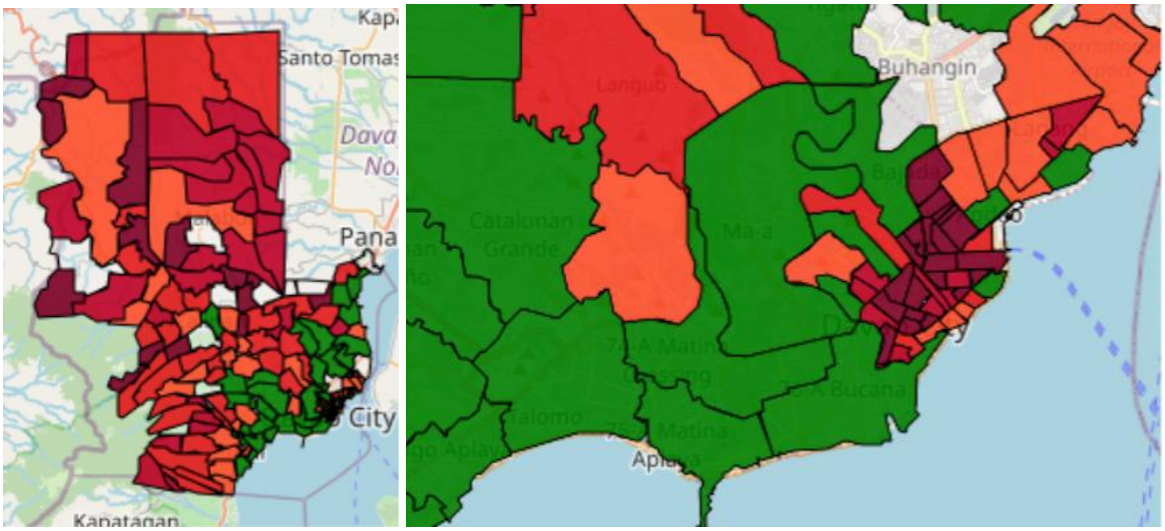

Figure 18: Heatmap displayed over map of Davao barangays

Based on the heatmap in figure 17, the findings show that barangays found inside Talomo and Buhangin Districts to be more effective in Anemia Tetanus Management, while majority of the barangays are found in the northern parts of Davao City. From the data, Talomo (Pob.) has the highest number of cases spanning from 2020 to 2024 whereas Barangay 3-A has the lowest number of cases.

Table 11: *Barangay Talomo and Barangay 3-A's number of cases from 2020-2024*

Barangay	Year	2 doses of Td vaccination	3+ doses of Td vaccination	CBC / Hgb & Hct Count	Anemia Diagnosis	Iron Supplementation	Completed Iron + Folic Acid	Health Counseling
Talomo	2020	454	367	769	155	128	956	976
	2021	459	307	765	116	108	1093	1096
	2022	359	330	973	256	89	1154	1156
	2023	382	362	1099	316	113	1172	1172
	2024	276	352	862	169	93	977	977

Barangay	Year	2 doses of Td vaccination	3+ doses of Td vaccination	CBC / Hgb & Hct Count	Anemia Diagnosis	Iron Supplementation	Completed Iron + Folic Acid	Health Counseling
3-A	2020	2	1	3	2	1	2	2
	2021	5	1	7	2	2	6	6
	2022	2	1	7	6	2	11	11
	2023	4	3	6	1	0	13	13
	2024	11	4	15	7	0	12	12

4.5.2 Birth Facilities

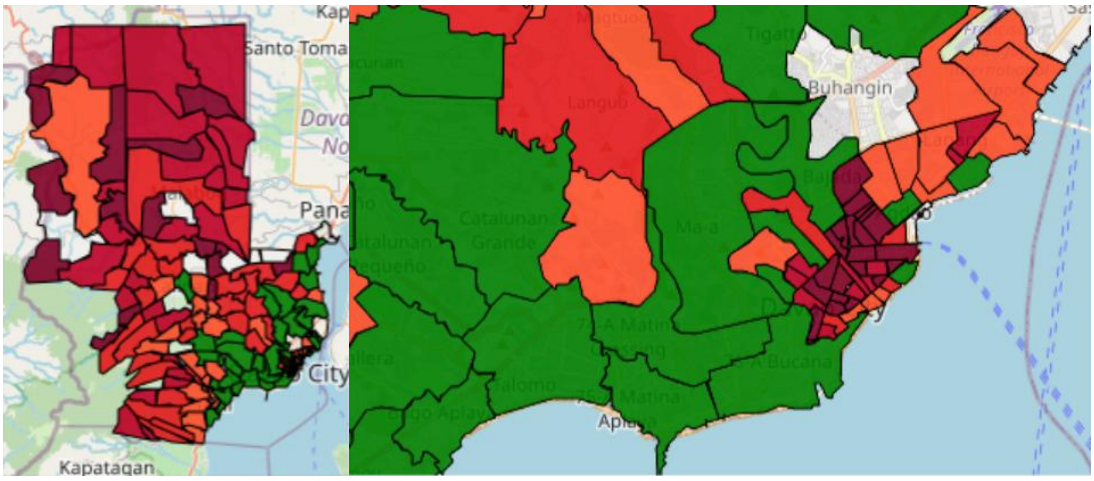

Figure 19: Overall and close-up birth facilities heatmap of Davao barangays

Birth Facilities refer to the number of birth facilities found in each barangay in Davao City. Red represents lower values whereas green represents higher values. The total value is calculated by average as the number spans between 2020 to 2024.

From the heatmap, barangay Ma-a has the highest number of cases whereas barangay 13-B has the lowest number of cases. The tables are split into two parts for easier readability.

Table 12.a: *Barangay Ma-a table of cases*

Barangay	Year	Private Deliveries	Gov't Health Facilities	Gov't Hospitals	Gov't Hosp. Normal Delivery	Gov't Hosp. Operative	RHU/ MHC/ BHS	Private Facilities
Ma-a	2020	488	611	611	440	171	0	636
	2021	403	442	442	263	179	0	560
	2022	392	619	618	447	171	1	623
	2023	404	741	741	529	212	0	547
	2024	301	595	595	412	183	0	345

Table 12.b: *Barangay Ma-a table of cases*

Barangay	Year	Private Hosp.	Private Birthing Homes	Other/Unknown Deliveries	Private (13.1.2)	Facility Delivery by Age	Normal Vaginal Delivery
Ma-a	2020	264	372	0	304	1248	793
	2021	264	296	0	254	979	512
	2022	276	347	1	271	1242	962
	2023	238	309	0	0	1288	977
	2024	141	204	0	0	940	695

Table 13.a: *Barangay 13-B table of cases*

Barangay	Year	Private Deliveries	Gov't Health Facilities	Gov't Hospitals	Gov't Hosp. Normal Delivery	Gov't Hosp. Operative	RHU/ MHC/ BHS	Private Facilities
13-B	2020	0	2	2	2	0	0	8
	2021	0	3	3	3	0	0	4
	2022	0		2	2			2

			2			0	0	
	2023	0		3	2			1
			3			1	0	
	2024	0		6	3			2
			6			3	0	

Table 13.b: *Barangay 13-B table of cases*

Barangay	Year	Private Hosp.	Private Birthing Homes	Other/Unknown Deliveries	Private (13.1.2)	Facility Delivery by Age	Normal Vaginal Delivery
13-B	2020	4	4	0	4	10	10
	2021	1	3	0	1	7	7
	2022	1	1	0	1	4	3
	2023	0	1	0	0	4	3
	2024	0	2	0	0	8	5

4.5.3 Health Districts

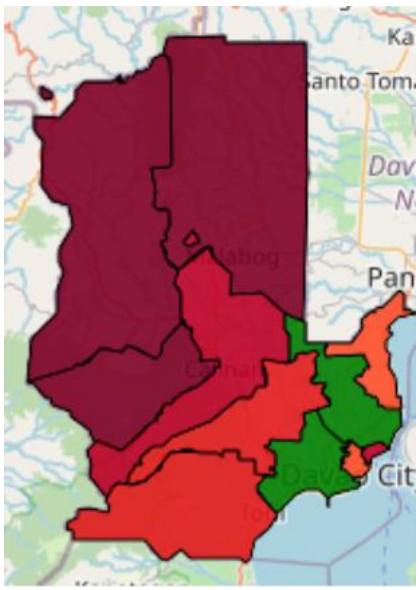

Figure 20: Heatmap of population in relation to medical personnel and neonatal deaths

Health Districts refer to the actual health districts registered by the CHO of Davao City. The dataset covers 18 health districts; however, some districts are merged to fit in the trace map,

particularly Talomo, Toril, and Poblacion Districts. This dataset is considered relative to the model as it provides the demographic framework upon which the model's calculations are based.

The heatmap is calculated based on the population found in the district in relation to the number of medical personnel and neonatal deaths. According to the heatmap, Talomo District has the highest number of populations in a span of five years whereas Baguio District has the lowest number of populations.

The table listed below for Talomo District consists of 3 parts, namely Talomo Central, Talomo North, and Talomo South. Furthermore, the tables are split for easier readability.

Table 14.a*: Talomo district number of medical personnel and neonatal deaths*

District	**Year**	**Projected Population**	**No. Of Barangays**	**Doctors**	**Dentists**	**Nurses**	**Midwives**	**Medical Technologists**
Talomo Central	2020	129474	5	1	1	3	3	1
	2021	132323	5	2	4	4	3	1
	2022	135236	5	3	2	4	5	1
	2023	137492	5	3	2	4	3	2
	2024	138864	5	3	1	2	5	2
Talomo North	2020	168802	1	2	1	0	5	1
	2021	172518	1	2	1	5	4	2
	2022	176315	1	4	2	3	4	2
	2023	152544	1	4	2	11	6	2
	2024	154055	1	4	0	2	0	2

Talomo South	2020	167508	10	1	1	2	5	1
	2021	171196	10	2	1	4	5	22
	2022	174964	10	0	0	0	0	0
	2023	166757	10	3	2	6	5	4
	2024	168450	10	3	1	6	5	11

Table 14.b: *Talomo district number of medical personnel and neonatal deaths*

District	**Year**	**Nutrionists / Dieticians**	**Neonatal Deaths (M)**	**Neonatal Deaths (F)**	**Neonatal Deaths (Total)**
Talomo Central	2020	1	1	2	3
	2021	1	1	0	1
	2022	1	0	1	1
	2023	1	27	12	39
	2024	1	2	4	6
Talomo North	2020	1	1	2	3
	2021	1	0	0	0
	2022	1	4	1	5
	2023	1	21	13	34
	2024	1	10	4	14
Talomo South	2020	2	0	0	0
	2021	2	0	0	0
	2022	0	0	0	0
	2023	2	16	8	24
	2024	2	4	8	12

Table 15.a: *Baguio district number of medical personnel and neonatal deaths*

District	Year	Projected Population	No. Of Barangays	Doctors	Dentists	Nurses	Midwives	Medical Technologists
Baguio District	2020	37689	8	0	0	0	3	0
	2021	38520	8	0	0	0	3	0
	2022	39368	8	1	1	0	2	0
	2023	39970	8	1	1	0	2	0
	2024	40341	8	1	1	4	3	0

Table 15.b: *Baguio district number of medical personnel and neonatal deaths*

District	Year	Nutrionists / Dieticians	Neonatal Deaths (M)	Neonatal Deaths (F)	Neonatal Deaths (Total)
Baguio District	2020	0	3	0	3
	2021	0	0	0	0
	2022	0	1	0	1
	2023	0	5	2	7
	2024	0	0	1	1

4.5.4 Post Partum Care and Natality

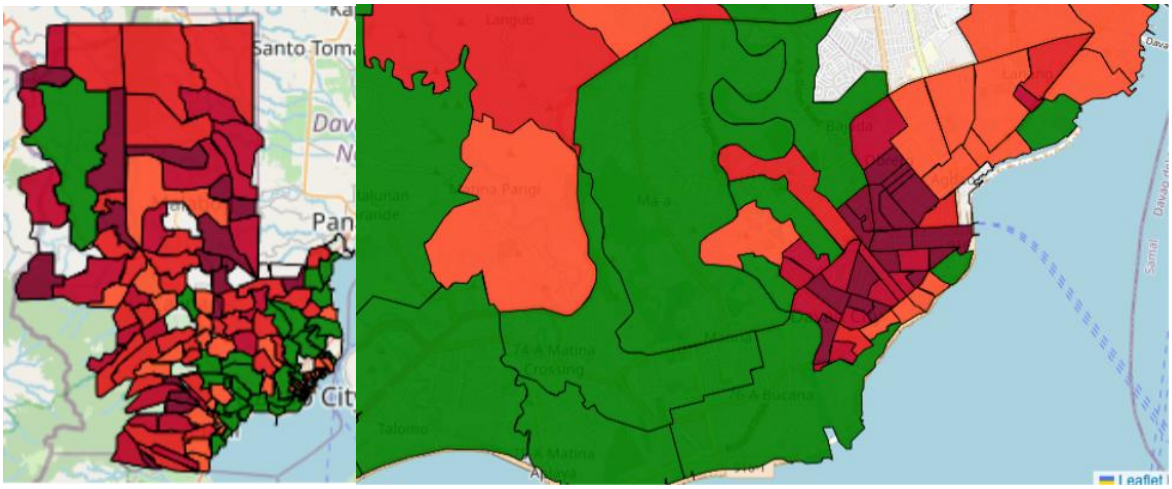

Figure 21: Heatmap of postpartum care and natality cases

Based on the heatmap, Ma-a has the highest number of cases in terms of Post Partum Care and Natality whereas Barangay 13-B has the lowest number of cases. The tables listed below are split into two parts for easier readability.

Table 16.a: *Barangay Ma-a number of Post Partum Care and Natality cases*

Barangay	Year	>2 Postpartum Visits	Breast feeding Initiated	Vitamin A Supplementation	Iodine Given	Health Counseling	Expected PP Clinic Visit	Quality PP Care
Ma-a	2020	1224	1136	1214	115	1260	1260	1260
	2021	1057	1064	1057	74	1064	1064	1064
	2022	1257	937	1257	740	1259	1259	938
	2023	1293	1299	1293	129	1299	0	1292
	2024	934	939	934	0	934	0	931

Table 16.b: *Barangay Ma-a number of Post Partum Care and Natality cases*

Barangay	Year	Hemoglobin Test after Delivery	Delivered 3 Months Ago	Completed 3mo Iron Supplement	FP Counseling
Ma-a	2020	314	1291	1150	1205
	2021	101	663	861	1064
	2022	390	1170	1170	1249
	2023	764	1308	1262	1293
	2024	934	1029	894	934

4.5.5 Pregnancy Outcomes

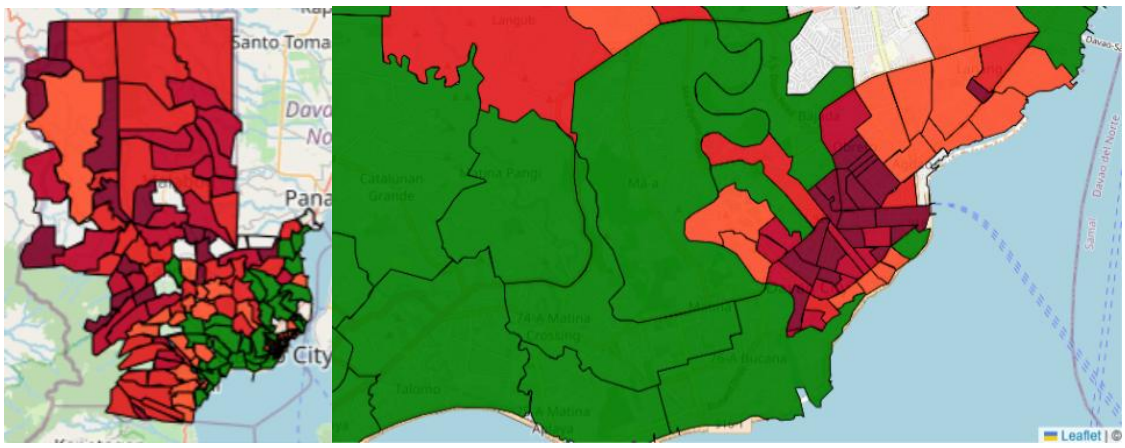

Figure 22: Heatmap of pregnancy outcomes

Pregnancy outcomes refer to cases of Abortion, Stillbirths, Live-Deliveries, Full-Term Births, and the number of doctors present in maternal care.

Based on the heat map, Ma-a has the highest number of cases while Dalagdag has the lowest number of cases.

Table 17*: Barangay Ma-a cases of pregnancy outcomes*

Barangay	Year	Abortion Cases	Stillbirths	Live Deliveries	Full-Term Births	Doctors
Ma-a	2020	0	0	1266	1258	891
	2021	0	0	1064	1023	753
	2022	0	1	1257	1251	849
	2023	0	1	1299	1298	954
	2024	0	5	939	938	709

Table 18*: Barangay Dalagdag cases of pregnancy outcomes*

Barangay	Year	Abortion Cases	Stillbirths	Live Deliveries	Full-Term Births	Doctors
Dalagdag	2020	0	0	12	12	3
	2021	0	0	18	18	3
	2022	0	1	9	9	7
	2023	2	2	23	23	8
	2024	0	0	18	18	9

4.5.6 Prenatal Care Interventions

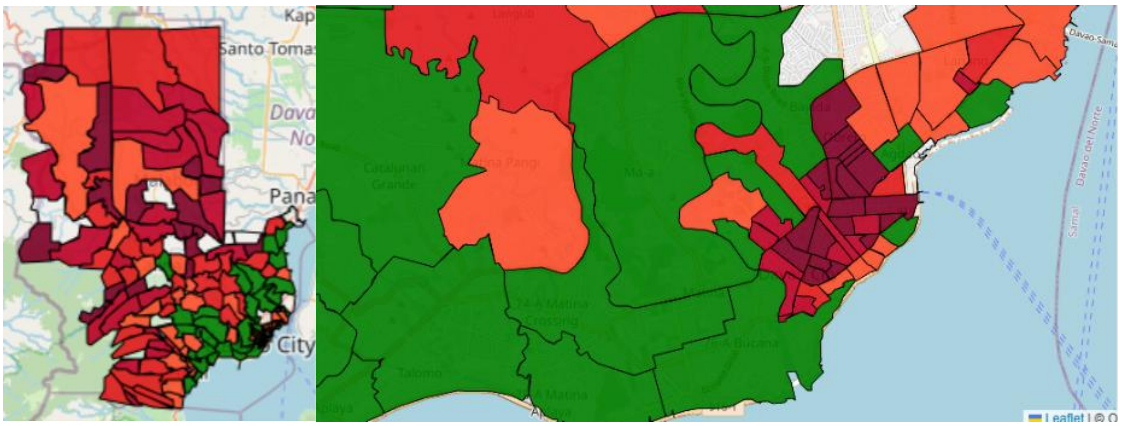

Figure 23: Heatmap of prenatal care interventions

Prenatal Care Interventions are medical services provided for pregnant women to monitor the status of the baby to prevent complications to the baby. The dataset consists of "With Birth Plan", Iodine Capules, Deworming Tablet, 1st Trimester Visit (Delivered), Completed Iron + Folic Acid (Delivered), Counselling (Delivered), and Completed Calcium Carbonate.

Based on the heatmap, Talomo (Pob.) has the highest number of cases while Barangay 3-A has the lowest number of cases.

Table 19*: Talomo (Pob.) number of prenatal care intervention cases*

Barangay	**Year**	**With Birth Plan**	**Iodine Capsules**	**Deworming Tablet**	**1st Trimester Visit (Delivered)**	**Completed Iron + Folic Acid (Delivered)**	**Counseling (Delivered)**	**Completed Calcium Carbonate**
Talomo (Pob.)	2020	1041	92	383	884	956	976	942
	2021	858	109	192	1090	1093	1096	687
	2022	999	0	199	728	1154	1156	840
	2023	1098	0	89	0	0	1172	1127
	2024	914	0	179	0	0	977	880

Table 20: *Barangay 3-A number of prenatal care intervention cases*

Barangay	**Year**	**With Birth Plan**	**Iodine Capsules**	**Deworming Tablet**	**1st Trimester Visit (Delivered)**	**Completed Iron + Folic Acid (Delivered)**	**Counseling (Delivered)**	**Completed Calcium Carbonate**
3-A	2020	4	2	1	2	2	2	2
	2021	7	5	2	6	6	6	6
	2022	7	3	1	8	11	11	11

	2023	6	6	3	0	0	13	13
	2024	15	11	13	0	0	12	12

4.5.7 Prenatal Visits

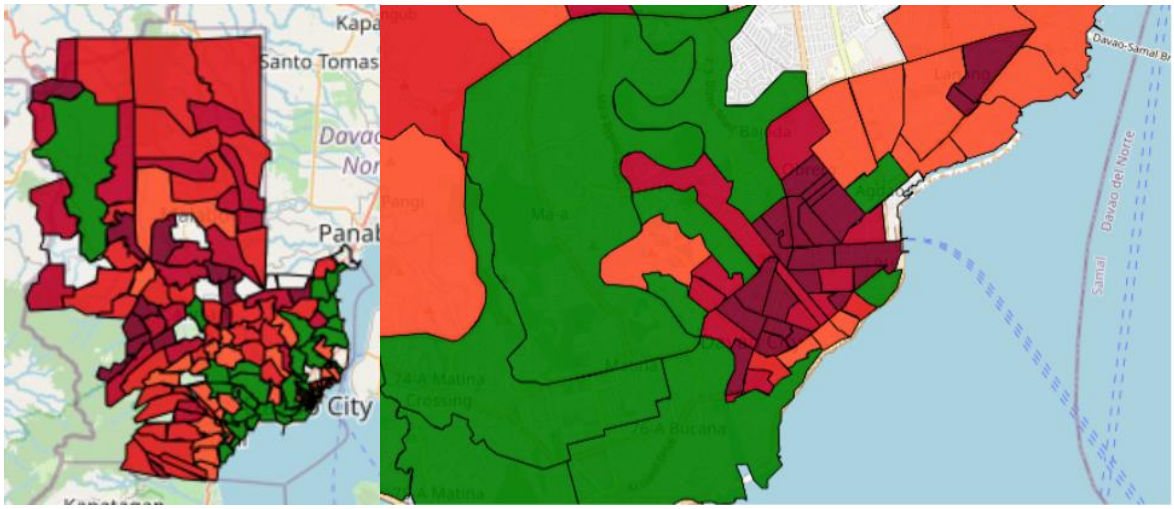

Figure 24: Heatmap of cases of prenatal visits

Prenatal Visits refer to the number of pregnant women visit the birth facilities in the span of 5 years. Based on the heatmap, Ma-a has the highest number of cases whereas Barangay 3-A has the lowest number of cases.

Table 21: *Barangay Ma-a cases of prenatal visits*

Barangay	Year	Total 1st Prenatal Visits	1st Trimester Visits	2nd Trimester Visits	3rd Trimester Visits
Ma-a	2020	1171	1003	120	48
	2021	869	702	97	70
	2022	1309	1135	101	73
	2023	1123	948	113	62

	2024	853	620	169	64

Table 22: *Barangay 3-A cases of prenatal visits*

Barangay	Year	Total 1st Prenatal Visits	1st Trimester Visits	2nd Trimester Visits	3rd Trimester Visits
3-A	2020	4	2	2	0
	2021	6	6	0	0
	2022	7	3	4	0
	2023	6	3	3	0
	2024	15	8	6	1

4.5.8 Screening and Testing

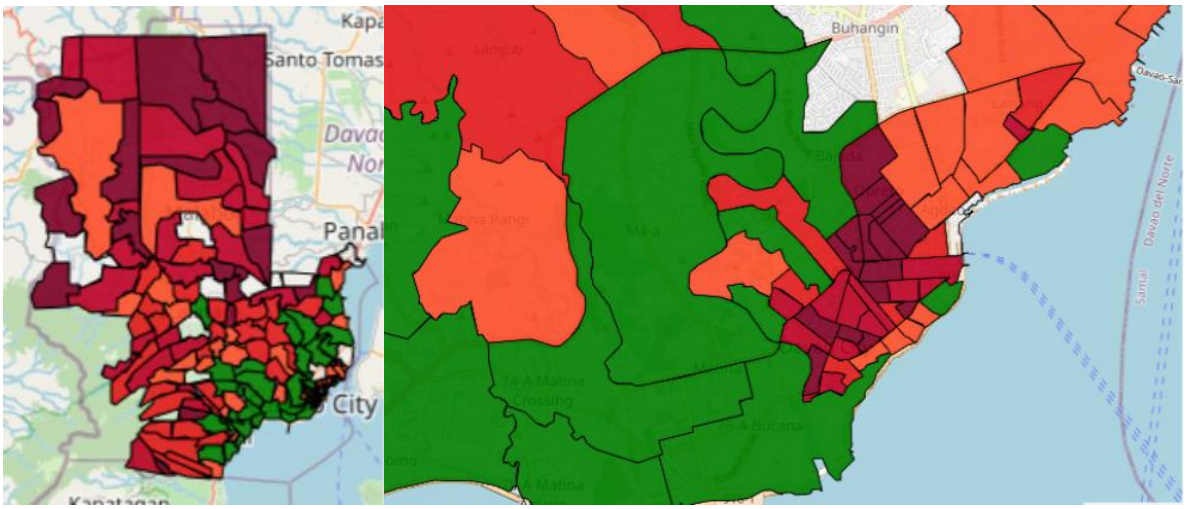

Figure 25: Heatmap of screening and testing in Davao City

Screening and Testing consists of various tests made for pregnant women in Davao City. Based on the heatmap, Talomo (Pob.) has the highest number of cases whereas Barangay 13-B has the lowest number of cases in the span of 5 years. The tables below are split into 2 parts for easier readability.

Table 23.a: *Talomo (Pob.) cases of screening and testing*

Barangay	Year	Syphilis Screening	Hepatitis B Testing	HIV Testing	Gestational Diabetes Screening
Talomo (Pob.)	2020	670	676	84	98
	2021	713	722	68	202
	2022	906	921	787	365
	2023	902	919	951	515
	2024	599	567	692	452

Table 23.b: *Talomo (Pob.) cases of screening and testing*

Barangay	Year	Hemoglobin Test	Blood Typing	Urinalysis	3 Basic Lab Exams
Talomo (Pob.)	2020	891	897	925	958
	2021	1100	1100	1100	1100
	2022	1149	1149	1149	1149
	2023	1172	1172	1173	1172
	2024	977	977	977	977

Table 24.a: *Barangay 13-B cases of screening and testing*

Barangay	Year	Syphilis Screening	Hepatitis B Testing	HIV Testing	Gestational Diabetes Screening
13-B	2020	0	0	0	0
	2021	0	0	0	0
	2022	2	3	0	0
	2023	6	6	6	0
	2024	8	8	8	0

Table 24.b: *Barangay 13-B cases of screening and testing*

Barangay	Year	Hemoglobin Test	Blood Typing	Urinalysis	3 Basic Lab Exams
13-B	2020	10	10	10	10
	2021	7	7	7	7
	2022	4	4	4	4
	2023	4	4	4	4
	2024	8	8	8	8

4.5.9 Skilled Birth Attendants

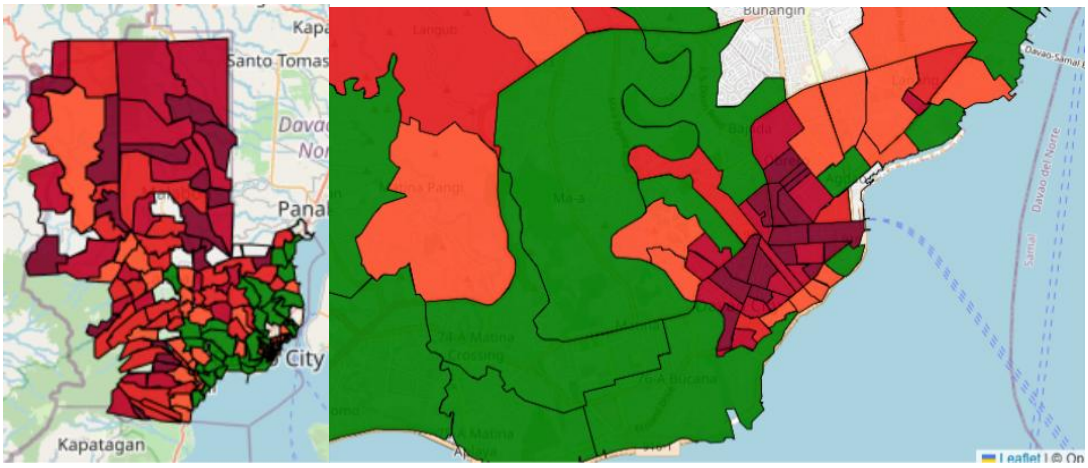

Figure 26: Heatmap of cases of skilled birth attendants

Skilled Birth Attendants refers to the number of visits seen by a doctor, the number of doctors from different types, and a number of various tests. Based on the heatmap, Ma-a has the highest number of cases while Barangay 13-B (Pob.) has the lowest number of cases. The tables below are split into three parts not only for easier readability but due to the number of columns present in the dataset.

Table 25.a: *Barangay Ma-a cases of skilled birth attendants*

Barangay	Year	Seen by Doctor	Gov't (Doctor)	Private (Doctor)	Private (3.2)	1st Trimester Visit (Delivered)	≥4 Prenatal Checkups	Lab Exams (Delivered)
Ma-a	2020	1147	659	488	537	1124	1030	0
	2021	879	476	403	395	818	862	0
	2022	1156	764	392	388	1137	1137	0
	2023	1147	743	404	405	1127	1127	0
	2024	856	555	301	288	775	775	0

Table 25.b: *Barangay Ma-a cases of skilled birth attendants*

Barangay	Year	Iron + Folic Acid (Delivered)	Counseling (Delivered)	Calcium Supplementation	Quality Prenatal Care	Abortion Cases	Operative Delivery	Attendant Birth
Ma-a	2020	1041	1064	1036	1036	0	129	1279
	2021	864	871	865	865	0	112	1063
	2022	1137	873	1137	1137	0	109	1258
	2023	0	1236	1120	1120	0	111	1300
	2024	0	856	775	775	0	66	944

Table 25.c: *Barangay Ma-a cases of skilled birth attendants*

Barangay	Year	Doctors (Attendant)	Gov't Doctors	Nurses (Attendant)	Midwives (Attendant)	Hilots	Others / Unknowns	Deliveries by SBA (Age Group)
Ma-a	2020	891	587	0	372	16	0	1248
	2021	753	499	0	296	14	0	980

	2022	849	578	19	375	15	0	1243
	2023	954	0	0	334	0	12	1288
	2024	709	0	0	231	0	4	940

Table 26.a: *Barangay 13-B (Pob.) cases of skilled birth attendants*

Barangay	**Year**	**Seen by Doctor**	**Gov't (Doctor)**	**Private (Doctor)**	**Private (3.2)**	**1st Trimester Visit (Delivered)**	**≥4 Prenatal Checkups**	**Lab Exams (Delivered)**
13-B (Pob.)	2020	10	10	0	3	10	10	0
	2021	7	7	0	5	7	7	0
	2022	4	4	0	0	4	4	0
	2023	4	4	0	1	4	4	0
	2024	8	8	0	8	8	8	0

Table 26.b: *Barangay 13-B (Pob.) cases of skilled birth attendants*

Barangay	**Year**	**Iron + Folic Acid (Delivered)**	**Counseling (Delivered)**	**Calcium Supplementation**	**Quality Prenatal Care**	**Abortion Cases**	**Operative Delivery**	**Attendant Birth**
13-B (Pob.)	2020	10	10	10	10	0	0	10
	2021	7	7	7	7	0	0	7
	2022	4	4	4	4	0	1	4
	2023	0	4	4	4	0	0	4
	2024	0	8	8	8	0	0	8

Table 26.c: *Barangay 13-B (Pob.) cases of skilled birth attendants*

Barangay	Year	Doctors (Attendant)	Gov't Doctors	Nurses (Attendant)	Midwives (Attendant)	Hilots	Others / Unknowns	Deliveries by SBA (Age Group)
13-B (Pob.)	2020	6	2	0	4	0	0	10
	2021	4	3	0	3	0	0	7
	2022	3	2	0	1	0	0	4
	2023	3	0	0	1	0	0	4
	2024	6	0	0	2	0	0	8

Among all the barangays in Davao City, Ma-a and Talomo (Pob.) are widely considered in all datasets to have the highest number of cases, indicating having the best performance in terms of maternal care interventions, whereas Barangay 3-A and 13-B are considered to have the lowest number of cases, indicating the worst performance.

With these findings, it proves and highlights the dashboard's effectiveness as a decision-support tool, enabling the identification of priority areas that require maternal care interventions.

4.6 Discussion of Findings

The results of the Causal Forest analysis provide several key insights into the dynamics of maternal care interventions and their effect on live birth rates in Davao City.

First, the ATE analysis clearly identifies that interventions directly related to the *time of birth* (rate_facility_delivery and rate_prof_attendant) have the largest causal impact, with point estimates nearly double that of any other intervention. The near-identical ATEs (0.0449 and 0.0445, respectively) and the consistent HTE predictors (nurses, midwives, doctors)

strongly suggest that these two variables are measuring the same core event: a professionally supervised, facility-based birth. It is logical to conclude that a birth attended by a professional is almost certain to have occurred within a health facility, making these two interventions highly collinear.

Second, the analysis confirms that *prenatal care* is a significant causal driver of the outcome. Interventions such as achieving at least four prenatal visits, starting visits in the first trimester, and conducting essential lab tests all demonstrated statistically significant positive effects. This underscores the importance of the entire continuum of care, from early pregnancy through to delivery.

Third, the Heterogeneous Treatment Effect (HTE) analysis confirms that the impact of these interventions is not uniform. The effectiveness of rate_facility_delivery, for example, appears to be significantly different in districts like Buhangin (lower effect) and Calinan (higher effect). While this analysis presents these findings as they are, it opens important avenues for future qualitative and quantitative investigation. The model suggests *that* heterogeneity exists; further research would be in-needed to understand *why*.

Finally, it is important to situate these findings within their methodological context. The Causal Forest framework offers a powerful, modern approach to estimate causal effects from observational data, and its robustness was confirmed through rigorous falsification testing. However, as with any statistical model, these results are best viewed as a sophisticated, data-driven tool for exploration and hypothesis generation. They highlight critical relationships and identify areas of significant heterogeneity, providing a strong foundation for further inquiry rather than serving as a sole, prescriptive basis for policy decisions.

The software developed in relation to this study proves to be a valuable tool for the CHO to conduct analysis on maternal care interventions. The usage of heatmaps to measure and visualize the severity of cases across each barangay or district provides critical insights

and gather smart problem-solving ideas into areas that require immediate attention and resource allocation.

5. Conclusion

This chapter synthesizes the entirety of the research, beginning with a summary of the study's objectives, methodology, and principal findings. It then draws a final conclusion, focusing on the methodological implications of the study and its role as an exploratory tool for public health. Finally, it outlines key limitations and proposes actionable recommendations for both the Davao City Health Office (CHO) and future academic research.

5.1 Summary of the Study

This thesis set out to address the persistent challenge of assessing maternal health interventions in Davao City, where health outcomes vary significantly across different barangays. The primary objective was to move beyond simple correlation and apply a robust causal inference framework to estimate the causal impact of specific antenatal interventions on maternal health outcomes using observational data.

To achieve this, the study utilized five years (2020-2024) of barangay-level aggregated data from the Davao City Routine Health Information System (RHIS). As detailed in Chapter 3, this administrative dataset required an extensive data processing pipeline to manage and clean 95 individual Excel files, which were formatted for manual data entry rather than statistical analysis.

The core of the methodology was the implementation of a Causal Forest, a machine learning algorithm designed to estimate both Average Treatment Effects (ATE) and, more importantly, Heterogeneous Treatment Effects (HTE). This approach allowed the study to not

only quantify the average impact of 21 different interventions but also to identify *how* and *where* these effects vary.

The quantitative results, presented in Chapter 4, yielded several key findings:

1. **Impact of Birth-Time Interventions:** Interventions directly related to the time of birth—specifically rate_facility_delivery (ATE: 0.0449) and rate_prof_attendant (ATE: 0.0445)—demonstrated the largest statistically significant causal impact on increasing the live birth rate.
2. **Impact of Prenatal Care:** A suite of prenatal interventions, including rate_4_prenatal_visits (ATE: 0.0253) and rate_visit_1st_tri (ATE: 0.0207), also showed significant positive causal effects, underscoring the importance of the entire continuum of care.
3. **Heterogeneity of Effects:** The HTE analysis confirmed that the impact of these interventions is not uniform across the city. The effectiveness of rate_facility_delivery was found to be significantly lower in districts such as Buhangin, Paquibato, and Tugbok, and significantly higher in Calinan.
4. **Predictors of Heterogeneity:** The Causal Forest model identified that local healthcare staffing levels—primarily the number of nurses (num_nurses) and midwives (num_midwives)—were the most important predictors in explaining this variation in treatment effect.

Finally, the study's causal claims were validated through a series of falsification tests, which confirmed that the model was not identifying spurious relationships.

5.2 Principal Conclusion

The principal conclusion of this thesis is not a simple policy prescription, but rather a methodological one: the Causal Forest framework is a highly suitable and powerful exploratory tool for generating causal hypotheses from complex, observational public health data.

In public health, it is often unethical or logistically impossible to conduct randomized controlled trials (RCTs) to determine, for example, which barangays should receive fewer nurses or less access to facility-based deliveries. This study demonstrates a robust alternative. By leveraging the natural variation present in the CHO's existing administrative data, the Causal Forest model can estimate causal effects, effectively serving as a data-driven "signpost" for policymakers (Zheng & Yin, 2023).

The findings from Chapter 4 should not be interpreted as a final, prescriptive basis for reallocating resources. Rather, they should be seen as a sophisticated tool for hypothesis generation (Kakimoto et al., 2022). The model's identification of Buhangin and Paquibato as areas where the positive effect of facility delivery is dampened does not mean the intervention is failing; it means the CHO now has a data-driven mandate to investigate *why*. The model tells policymakers *where to look* and *what covariates matter*, allowing for the efficient allocation of limited qualitative and operational research resources.

This study, therefore, serves as proof of concept, demonstrating a path from static administrative reporting to a dynamic, analytical decision-support system for the Davao City Health Office.

5.3 Limitations of the Study

While the Causal Forest methodology is robust, the study's conclusions must be contextualized by several key limitations:

1. **Ecological Fallacy:** The analysis was conducted at the aggregated barangay-month level, not on individual patient data. Therefore, conclusions drawn about barangays cannot be definitively applied to individuals within them.
2. **Data Infrastructure:** As identified in Chapter 3, the current RHIS data structure relies on manually intensive and error-prone consolidation from numerous files. This practical limitation not only made data processing a significant hurdle but also limits the potential for real-time analysis, which would be necessary for a truly dynamic decision-support system.
3. **Omitted Variable Bias:** While Causal Forest is designed to be robust against confounding, the model is limited to the variables present in the RHIS dataset. Unobserved confounders—such as local socio-economic conditions, community-level trust in the health system, or the prevalence and quality of private health facilities—were not included and could influence the estimated effects.
4. **Geospatial Validation:** The causal estimates from the Causal Forest model are robust for assessing barangay-level treatment effects, however the geospatial input data was not validated for autocorrelation. Thus, formal spatial statistics (e.g., Global Moran's I) were not applied to these rates. The geographic patterns visualized in the heatmaps are indicative but not statistically confirmed as true spatial clusters, which means the model may not account for all spatial confounding effects.

5.4 Recommendations

Based on the study's findings and limitations, the following recommendations are proposed for the Davao City Health Office and for future research.

5.4.1 Recommendations for the Davao City Health Office (CHO)

1. **Adopt a Targeted, Mixed-Methods Approach for Policy:**
    - **Action:** Use the HTE findings from this study as a guide for targeted operational research.
    - **Rationale:** Instead of immediately reallocating personnel, the CHO should commission qualitative studies (e.g., focus groups, key informant interviews) in the districts identified by the model (Buhangin, Paquibato, Tugbok) to understand the on-the-ground barriers (e.g., transportation access, facility costs, cultural preferences, perceived quality of care) that may be suppressing the positive effects of maternal interventions. The model provides the "what" and "where"; qualitative research is needed to provide the "why."
2. **Invest in Data Infrastructure Modernization:**
    - **Action:** Transition from the current Excel-based reporting system to a centralized, relational database, similar to the PostgreSQL framework developed for this thesis.
    - **Rationale:** This single recommendation would resolve the most significant practical limitation of this study. A centralized database would eliminate redundant data entry, reduce errors, and, when paired with automated analysis scripts, empower the CHO to run causal analyses regularly. This moves the city's health monitoring capability from static historical reporting to dynamic, near-real-time decision support.

5.4.2 Recommendations for Further Study

1. **Methodological Expansion:**
    - **Action:** Apply the Causal Forest framework to other critical health outcomes available in the RHIS dataset, such as neonatal mortality rates, stillbirth rates, or the incidence of specific maternal morbidities.
    - **Rationale:** This study focused on live birth rate as the primary outcome. Expanding the analysis would provide a more holistic picture of intervention effectiveness across the full spectrum of maternal and child health.
2. **Incorporate Spatio-Temporal Features:**
    - **Action:** Future models should more explicitly account for spatial and temporal autocorrelation.
    - **Rationale:** While this study included district and month as covariates, more advanced techniques, such as spatially-aware Causal Forests, could yield even more precise estimates by modeling how intervention effects "spill over" into adjacent barangays
3. **Promote Causal Inference in Philippine Public Health:**
    - **Action:** Encourage the adoption and application of this causal framework to other observational health datasets within the Philippines.
    - **Rationale:** This thesis serves as a case study for moving beyond correlational analysis. By applying robust causal machine learning methods, public health researchers and policymakers can gain a much deeper and more reliable understanding of "what works" from the administrative data they already collect.

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APPENDIX A

CONSULTATION LETTER

MAPÚA
MALAYAN COLLEGES
MINDANAO

MS. MELODY C. HERRERA
Acting Department Head
Human Resource Management Office
2nd Floor, City Hall, Davao City

Dear Ms. Herrera,

Good day! We are 3rd year students of Mapua Malayan Colleges Mindanao under the program BS Computer Science. We are currently working with our thesis project entitled "Integrated Machine Learning System for Disease Pattern Analysis and Outbreak Detection in Davao City Barangays".

In this thesis project, we aim to develop a system for predictive analysis of data collected by operations, and to provide insights in the improvement of statistics-based decision making.

We would like to request a consultation with members who are willing to provide insights with regards to our thesis project. This would greatly benefit our approach to create a robust product and to feasibly accomplish this project.

We greatly appreciate your time and guidance. Hoping for your kind consideration.

Sincerely,

RAPHAEL MICO BARRIOS
3rd Year BS Computer Science Student

JC-REY CATA-AL
3rd Year BS Computer Science Student

HUMAN WALTER OLO
3rd Year BS Computer Science Student